

Exact selectors, character energy, and integral return for the Erdős–Straus equation

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12 September 2026

Abstract

The bounded divisor formulation of the Erdős–Straus equation admits exact arithmetic, character-theoretic and geometric descriptions. We prove completeness of the first-half denominator selector and give its inverse coordinate maps. Inversion parity yields an exact collision-energy minimum for a divisor distribution avoiding all solution targets; resolving this minimum by quadratic characters strengthens the resulting existence test. The original arithmetic gate is also the exact order of a specified affine cocycle. An integral three-coordinate change identifies its cyclic action with a hexagonal quotient, retaining the third coordinate and index-three gluing. The homogeneous cyclotomic quartic is the same quadratic form on positive squares, with a complete description of its dihedral square returns. Local compactification maps are given using actual branch-cycle lengths. These results do not establish universal Erdős–Straus existence.

Contents

1	Research lineage and conventions	3
2	The complete first-half selector and its cyclotomic relations	3
2.1	The denominator bound, divisor maps and exact code	3
2.2	Endpoint forcing and the corrected generalized carrier	6
2.3	Cyclotomic order, infinitude, coverage and a smaller cofactor	8
2.4	Necessary nonresidue factors and the exact character target	10
2.5	Five exact regressions and their scopes	12
3	Exact character-resolved collision minima and integral ES return	14
3.1	The complete shell, exponent box, and ordered return	14
3.2	Multiplicities, parity, and character cells	16
3.3	The exact minimum, including all impossible cases	17
3.4	Exact refinement and preservation of certificates	18
3.5	The complete numerical witness at $p = 1201$	19
3.6	Finite verification and the boundary of the result	20
4	The original gate in an integral three-coordinate system	21
5	JT’s quartic and the positive-square locus of the hexagon	23
6	Local maps of the completed affine covers	23

7	The marked cubic-surface contraction and the toric hexagon	25
7.1	The marked surfaces, contraction and complete inverse fibres	25
7.2	All twenty-seven line classes and their contracted images	27
7.3	A six-cycle realized by Cremona and a marked configuration map	27
7.4	Torus coordinates, the exact fan, and the cusp-side convention	28

1 Research lineage and conventions

The Boolean, positivity, cyclotomic and prime-character directions in the joint investigation are credited to JT as identified in the supplied dialogue. The maintainer's programme asks for exact maps linking those arithmetic conditions to the geometric structures, without discarding coordinates or scales. Assistant-derived arguments are recorded as such, not retroactively attributed to a human who requested their investigation. The source locators and hashes in `../PROVENANCE.md` distinguish the complete dialogue, recovered source packages, and the newly written proofs.

The type-I/type-II arithmetic has its literature antecedent in Elsholtz and Tao [3]; the received papers preserve their more detailed source-specific references. The integral monodromy used below is taken from the displayed matrices of the circulated period manuscript [4], through the received torsion-cover note. Only those explicit matrices enter these algebraic proofs. The global sphere claim of that manuscript is not a premise. Permutation triples and branched covers have the classical context described by Sijsling and Voight [5].

Each section declares its own coordinates: in particular a primitive tuple's scale D and an affine system's level D are named locally and are not silently identified. Ordered denominators, channel marks, divisor exponents and branch origins are retained. Finite computations are distinguished from the general written proofs.

2 The complete first-half selector and its cyclotomic relations

The source dialogue is JT and ChatGPT, *Boolean product formulation*, 9–11 September 2026, exported 12 September 2026 ([shared source dialogue](#)). The preserved transcript has SHA-256 `19d54ae24d712598c14f222ee81eb19f192c9e944fbbfe5009f91780709792f3`. The locators below refer to the one-based lines of that export. JT supplies the Boolean-product, cyclotomic, generalized-congruence and adversarial directions; the following arguments state the mathematical claims with their exact scopes. They make no blanket claim of novelty. In particular no universal Erdős–Straus theorem or counterexample is asserted. The first-half proof and the explicit complete permutation fibres below fill out the relation between the retained full-range code and the dialogue's first-half code.

Write $\text{ES}(p)$ for the existence of positive integers x, y, z such that $4/p = 1/x + 1/y + 1/z$. Prime factorization, quadratic reciprocity with its supplementary laws, and cyclicity of the multiplicative group of a finite field are used in their usual classical forms. Every additional arithmetic implication in this section is proved here.

2.1 The denominator bound, divisor maps and exact code

Theorem 2.1 (First-half denominator bound). *Let p be prime and $p \equiv 1 \pmod{4}$. The minimum denominator in every positive integer Erdős–Straus solution lies strictly between $p/4$ and $p/2$.*

Proof. Retain a permutation of the original coordinates and name its ordered values $x \leq y \leq z$. Positivity gives $1/x < 4/p$, so $x > p/4$. Suppose $x > p/2$; equality is impossible because p is odd. Then

$$\frac{4}{p} \leq \frac{1}{x} + \frac{2}{y} < \frac{2}{p} + \frac{2}{y},$$

so $y < p$. Neither x nor y is divisible by p . The integer equation $4xyz = p(xy + xz + yz)$ gives $p \mid z$. Write $z = pk$. The original equality becomes

$$(4k - 1)xy = pk(x + y).$$

Thus $t = (4k - 1)/p$ is a positive integer, $t \equiv 3 \pmod{4}$, $\gcd(t, k) = 1$, and $txy = k(x + y)$. Write $x = g\alpha, y = g\beta$, where $g = \gcd(x, y)$ and $\gcd(\alpha, \beta) = 1$. Then

$$tg\alpha\beta = k(\alpha + \beta), \quad \gcd(\alpha\beta, \alpha + \beta) = 1.$$

Hence $k = \alpha\beta D$ for an integer $D > 0$, and

$$tg = D(\alpha + \beta), \quad \gcd(t, D) = 1, \quad pt = 4\alpha\beta D - 1.$$

The inequality $x > p/2$ implies

$$2D\alpha(\alpha + \beta) > 4\alpha\beta D - 1, \quad 2D\alpha(\alpha - \beta) > -1.$$

The last left side is an integer, so it is nonnegative and $\alpha \geq \beta$. Applying the same argument to $y > p/2$ gives $\beta \geq \alpha$. Coprimality forces $\alpha = \beta = 1$. Now $tg = 2D$ and $\gcd(t, D) = 1$ give $t \mid 2$, contrary to $t \equiv 3 \pmod{4}$. The retained permutation recovers the original coordinate order. $\square$

Fix henceforth a prime $p = 12h + 1$, $h \geq 1$, and put

$$a = 3h + j + 1, \quad R = 4a - p = 4j + 3, \quad S = pa, \quad 0 \leq j < 3h.$$

Thus $p/4 < a < p/2$, $3 \leq R \leq p - 2$, and $\gcd(R, pa) = 1$. Indeed $a < p$ is prime to p , a common divisor of a, R divides p , and a common divisor of p, R divides $4a$; also R is odd. Define

$$E_a = \{u > 0 : u \mid a^2, R \mid 4u + 1\}, \quad M_a = \{u > 0 : u \mid a^2, R \mid 4u + p\}.$$

The middle condition is equivalently $R \mid u + a$.

Theorem 2.2 (Original ordered solutions and the bounded divisor code). *For $u \mid a^2$, set*

$$g = \gcd(a, u), \quad A = a/g, \quad B = u/g, \quad D = g^2/u.$$

These are positive integers and define a bijection

$$\text{Div}(a^2) \longleftrightarrow \{(A, B, D) > 0 : a = ABD, \gcd(A, B) = 1\}, \quad u = B^2 D.$$

For $\varepsilon \in \{0, 1\}$, its hit condition is exactly

$$R \mid 4u + p^\varepsilon \iff R \mid A + p^{1-\varepsilon} B.$$

On that set, define

$$C = \frac{A + p^{1-\varepsilon} B}{R}, \quad (x, y, z) = (ABD, p^\varepsilon ACD, pBCD). \quad (\text{BP1})$$

Then $C > 0$ is integral, $4ABCD = A + p^{1-\varepsilon} B + pC$, and (BP1) is an original ordered solution. Put

$$M_H = 6h, \quad J_H = 3h, \quad \Lambda_H = 36h^2, \quad N_H = 216h^3 = (p - 1)^3/8.$$

The map

$$c_H = (A - 1)\Lambda_H + (p - 1)j + 2(B - 1) + \varepsilon \quad (\text{BP2})$$

is a bijection from the rectangle $1 \leq A, B \leq M_H$, $0 \leq j < J_H$, $\varepsilon \in \{0, 1\}$ to $0 \leq c_H < N_H$. On its decoded coordinates set

$$\chi_{H,p}(c_H) = \left\lfloor \frac{\gcd(4AB, p + 4j + 3)}{4AB} \right\rfloor \left\lfloor \frac{1}{\gcd(A, B)} \right\rfloor \left\lfloor \frac{\gcd(4j + 3, A + p^{1-\varepsilon} B)}{4j + 3} \right\rfloor.$$

Then

$$\text{ES}(p) \iff \exists 0 \leq c_H < N_H : \chi_{H,p}(c_H) = 1. \quad (\text{BP3})$$

Proof. Write $a = gA, u = gB$ with $\gcd(A, B) = 1$. The divisibility $u \mid a^2$ implies $B \mid gA^2$, hence $B \mid g$. Therefore $D = g/B$ is integral and $a = ABD, u = B^2D$. Conversely these equalities give $u \mid a^2$ with quotient A^2D , and $\gcd(a, u) = BD$. This proves both inverse compositions. Also $A, B \leq a \leq M_H$. Using $4ABD = p + R$, one has

$$A(4u + p^\varepsilon) \equiv pB + p^\varepsilon A = p^\varepsilon(A + p^{1-\varepsilon}B) \pmod{R}.$$

Both A and p are units modulo R , proving the biconditional. On its passing locus, multiplying the definition of C by $R = 4ABD - p$ gives the tuple equation. The sum of the three reciprocals in (BP1), over denominator $pABCD$, has numerator $pC + p^{1-\varepsilon}B + A = 4ABCD$.

For the code inverse divide c_H by Λ_H to recover $A - 1$ and remainder v , divide v by $p - 1$ to recover j and remainder r , then divide r by 2 to recover $B - 1$ and ε . The ranges of successive remainders are precisely the prescribed ranges. The gcd-floor factors equal the divisibility or coprimality bits because their quotients are positive and at most one. Thus a passing code gives (BP1).

For the converse, Theorem 2.1 selects a first denominator a in this interval, retaining its original position. For the other denominators define

$$d_y = Ry - S = Sy/z > 0, \quad d_z = Rz - S > 0, \quad d_y d_z = S^2.$$

Conversely a positive $d_y \mid S^2$ with $R \mid d_y + S$ determines $d_z = S^2/d_y$ and $y = (d_y + S)/R, z = (d_z + S)/R$; the second quotient is integral because $d_y \equiv -S$ is a unit modulo R . Since $p \nmid a$, the exponent $i = v_p(d_y)$ is 0, 1 or 2. For $i = 0$, set $\varepsilon = 0, u = a^2/d_y$; multiplying $d_y + pa \equiv 0$ by $4u/a$ gives $p(1 + 4u) \equiv 0$. For $i = 2$, set $\varepsilon = 0, u = d_y/p^2$; multiplication by $4/p$ gives the same congruence, with the final two original denominators exchanged in (BP1). For $i = 1$, write $d_y = pv$ and set $\varepsilon = 1, u = a^2/v$. Then $v \equiv -a$, hence $u \equiv -a$, giving the middle condition. All three cases therefore give a passing code. The exterior exchange is recorded, not identified with the original ordering. This proves (BP3). $\square$

Proposition 2.3 (Radix embedding and the full permutation fibres). *Retain the repository's full rectangle*

$$J_F = 6h, \quad M_F = 9h, \quad \Lambda_F = 108h^2, \quad N_F = 972h^3,$$

with the same shell formula and code $c_F = (A - 1)\Lambda_F + 18hj + 2(B - 1) + \varepsilon$. The full-range hits satisfying $a \leq 6h$ are bijective to the first-half hits by decoding and recoding the same four coordinates. The map preserves a, R, S, A, B, C, D, u , the ordered triple (BP1), and every predicate on these coordinates, and

$$c_F - c_H = 2(A - 1)\Lambda_H + 6hj. \tag{BP4}$$

All other full-range hits still have a computable image among first-half hits after retaining the coordinate permutation. More precisely, let $\mathcal{H}_{\min}$ consist of first-half hit codes whose first denominator is minimal and whose middle-channel lift has $y < z$. There is a surjection from full-range hits to $\mathcal{H}_{\min}$. For a fixed code in $\mathcal{H}_{\min}$, its entire inverse fibre is obtained by the following finite rule: take all distinct permutations (b, c, d) of its lifted triple, retain those with

$$3h + 1 \leq b \leq 9h, \quad v_p((4b - p)c - pb) \in \{0, 1\}, \tag{BP5}$$

and recover their full codes by the inverse in the proof. Thus the two existential selectors agree, although their full hit sets, shell labels and total counts need not agree.

Proof. For a full-range hit with $a \leq 6h$, the conditions $AB \mid a$ give $A, B \leq 6h$, and $j < 3h$. The hit tests on those same coordinates are identical. The two proved radix decoders are inverse, and the values $\Lambda_F = 3\Lambda_H, 18h - 12h = 6h$ give (BP4).

For the second statement lift a full code to its ordered triple. Choose its minimum denominator as first coordinate. Theorem 2.1 puts it in the first-half interval. The residual-divisor proof in Theorem 2.2 gives a unique canonical exterior orientation, with residual exponent 0 for the second denominator, or a middle orientation with both exponents 1. In the latter case choose the smaller of the final two denominators as second. They cannot be equal: equality would give $d_y = d_z = S$, hence $R \mid 2S$, and the unit condition would give $R \mid 2$, impossible. This yields a unique code in $\mathcal{H}_{\min}$. In the exterior case exactly one of the two residual exponents is 0 and the other 2, so its orientation is also unique. The minimum value and the multiset of remaining values are unique, even if a value occurs more than once; hence these rules give a unique resulting code.

For a retained permutation in (BP5), put $R_b = 4b - p$, $S_b = pb$ and $d_c = R_b c - S_b$. When $v_p(d_c) = 0$, recover $\varepsilon = 0, u = b^2/d_c$; when $v_p(d_c) = 1$, recover $\varepsilon = 1, u = pb^2/d_c$. The proof of Theorem 2.2 gives the positive integers A, B, C, D and the code with $j = b - 3h - 1$. Since $A, B \leq b \leq 9h$, it is in the full rectangle and its ordered lift is exactly (b, c, d) . Conversely every full code in the inverse fibre has that triple as one of the distinct permutations, its first coordinate is in the full interval, and its canonical residual exponent is 0 or 1. Thus (BP5) gives every point of the fibre and no duplicates. Surjectivity follows because each minimal first-half code has its identical full-radix lift. $\square$

Corollary 2.4 (Boolean products and the exact two-layer inclusion). *Define*

$$S(p) = 1 - \prod_{c=0}^{N_H-1} (1 - \chi_{H,p}(c)), \quad S_2(p) = 1 - \prod_{c=0}^{2\Lambda_H-1} (1 - \chi_{H,p}(c)).$$

Then $S_2(p) \leq S(p)$, $S(p) = 1$ if and only if $\text{ES}(p)$, and $S_2(p) = 1$ if and only if a first-half hit has $A \in \{1, 2\}$. For any prime set $\mathcal{P}$ define its infinite Boolean product as the infimum of its finite initial products. Then

$$\prod_{p \in \mathcal{P}} S(p) = 1 \iff (\forall p \in \mathcal{P}) \text{ES}(p).$$

Proof. Each complement factor is a bit. Its finite product is one exactly when every hit bit vanishes. The radix intervals for fixed A are $[(A-1)\Lambda_H, A\Lambda_H - 1]$, so the first two are exactly the displayed smaller range. Inclusion gives the inequality. The finite initial products form a nonincreasing sequence of bits; one zero factor makes every later product zero, and absent such a factor every finite product is one. Apply (BP3). $\square$

These statements correspond to transcript lines 14–163, 180–387, 5203–5293 and 6465–6606. The repository’s previous exact radix map is Theorem `raw-first-half-code` in `raw_scale_hit_incidence.tex`; the broader full-range bound and code occur in `bounded_transport.tex` and `unary_boolean_transport.tex`.

2.2 Endpoint forcing and the corrected generalized carrier

Theorem 2.5 (The endpoint factor theorem). *Let $p \equiv 1 \pmod{24}$ be prime. An $A = 1$ hit with $R^2 \leq p$ and $B = 1$ or $D = 1$ exists if and only if $(p+1)/2$ or $p+4$ has a prime divisor congruent to 3 modulo 4. If q is the least such divisor of the first integer, the exterior code is $(p-1)(q-3)/4$. If it is the least such divisor of the second integer, the middle code is $(p+1)(q-1)/4$.*

Proof. Both integers are odd and 1 modulo 4. If either has a prime factor 3 modulo 4, the number of such prime factors counted with multiplicity is positive and even. The least one q

therefore has square at most that integer. For $(p+1)/2$, this gives $q^2 < p$. For $p+4$, it gives $q^2 \leq p+4$; since $q^2 - p$ is divisible by 8, it follows that $q^2 \leq p$.

In the first case use

$$(R, A, B, D, \varepsilon) = (q, 1, 1, (p+q)/4, 0), \quad C = (p+1)/q.$$

In the second case use

$$(R, A, B, D, \varepsilon) = (q, 1, (p+q)/4, 1, 1), \quad C = (p+q+4)/(4q).$$

All these quantities are positive integers; in the second case oddness of q and $q \mid p+4$ give the last integrality. Substitution in (BP2) gives the stated codes, and the tuple equations prove the hits.

Conversely $A = B = 1$ makes the middle channel impossible because $R \nmid 2$; the exterior channel requires $R \mid p+1$. If $A = D = 1$, then $B = a$, and the middle channel requires $R \mid p+4$. The exterior channel would require $R \mid p^2+4$. A positive integer $R \equiv 3 \pmod{4}$ has a prime factor $q \equiv 3 \pmod{4}$. Such a prime cannot divide p^2+4 : division by the unit 4 would make -1 a square modulo q . A square root of -1 would have order 4 in a group of order $q-1 \equiv 2 \pmod{4}$, impossible. In each remaining case that prime $q \mid R$ divides $(p+1)/2$ or $p+4$. $\square$

The factor hypothesis is not universal: at $p = 193$, the two integers are the primes 97 and 197, both 1 modulo 4. Nevertheless $(A, B, C, D, \varepsilon) = (1, 5, 138, 10, 0)$ gives the code 200 and the solution (50, 1380, 1331700). This retains the actual scope of transcript lines 2570–2944 and its correction at 5149–5189.

Theorem 2.6 (Exact generalized carrier). *For $u > 0$ set $\kappa(u) = \prod_{\ell \in \mathbb{N}} \ell^{\lceil e/2 \rceil}$. Let $n > 0$ and $R > 0$, $R \equiv 3 \pmod{4}$. The system*

$$t \equiv 1 \pmod{12n}, \quad t \equiv -R \pmod{4\kappa(u)}, \quad t \equiv -R - 4u \pmod{4R} \quad (\text{BP6})$$

has an integer solution if and only if

$$\gcd(3n, \kappa(u)) \mid (R+1)/4, \quad \gcd(3n, R) \mid (R+4u+1)/4. \quad (\text{BP7})$$

Its complete solution set, when nonempty, is one class modulo $4 \operatorname{lcm}(3n, \kappa(u), R)$. For the exterior carrier retain $R \mid 4u+1$ and omit the third congruence; only the first divisibility in (BP7) remains. For a prescribed prime p , actual membership in the carrier, $R < p$, and the original common divisor must be retained.

Proof. Prime valuations give $u \mid a^2$ if and only if $\kappa(u) \mid a$, and $\kappa(u) \mid u$. With $4a = p + R$, these are exactly

$$u \mid a^2 \iff p \equiv -R \pmod{4\kappa(u)}, \quad R \mid a + u \iff p \equiv -R - 4u \pmod{4R}.$$

For a finite system of integer congruences, pairwise residue agreement modulo the corresponding gcds is necessary by subtraction. To prove sufficiency, for each prime select a modulus having maximal exponent of that prime. Its prescribed residue agrees with every other at all lower powers by the pairwise condition. Ordinary CRT over these distinct prime powers gives a solution. Differences of solutions are exactly multiples of the lcm. Applying this criterion to (BP6), the first two pair conditions are (BP7); the third is $\gcd(\kappa(u), R) \mid u$, automatic because $\kappa(u) \mid u$. The lcm gives the stated period. The same argument with two moduli gives the exterior statement.

For a class $b + m\mathbb{Z}$, intersection with an integer interval $[\alpha, \beta]$ occurs exactly when

$$\left\lceil \frac{\alpha - b}{m} \right\rceil \leq \left\lfloor \frac{\beta - b}{m} \right\rfloor.$$

At $\alpha = \beta = p$, this is precisely membership of p . Finally a single $u \mid a^2$ has the unique exponent vector $(e_\ell) \in \prod_{\ell \mid a} \{0, \dots, 2v_\ell(a)\}$. Reduction modulo different prime powers dividing R uses this same vector. Its complete fibre is the intersection of the local fibres on that box, not their Cartesian product. $\square$

At $n = 2$, (BP7) is exactly

$$2 \mid u \Rightarrow R \equiv 7 \pmod{8}, \quad 3 \mid u \Rightarrow R \equiv 2 \pmod{3}, \quad 3 \mid R \Rightarrow u \equiv 2 \pmod{3}.$$

Thus a modulo-8 condition cannot be removed: $R = 3, u = 2$ gives incompatible requirements $t \equiv 1 \pmod{24}$ and $t \equiv 5 \pmod{8}$. Also $R = 7, u = 1$ gives the compatible class $73 \pmod{168}$, which does not contain 193. These are the corrections of transcript lines 4740–5058 and the generalization at 6104–6158.

2.3 Cyclotomic order, infinitude, coverage and a smaller cofactor

Theorem 2.7 (Cyclotomic Euclid and complete-family coverage). *Let $m = 12n$, $n > 0$, and $P > 0$. Then*

$$\Phi_m(mP) > 1, \quad \gcd(mP, \Phi_m(mP)) = 1,$$

and every prime divisor of this value is 1 modulo m . There are infinitely many primes in that class. For each such prime p , there are exactly $\varphi(m)/2$ integers $1 \leq r \leq (p-1)/2$ with $p \mid \Phi_m(mr)$. Consequently, with $P_T = \prod_{r=1}^T \Phi_m(mr)$,

$$p \equiv 1 \pmod{m}, \quad p \leq 2T + 1 \quad \implies \quad p \mid P_T. \quad (\text{BP8})$$

In the Boolean convention of the preceding corollary,

$$\prod_{p \equiv 1 \pmod{m}} S(p) = \inf_{T \geq 1} \prod_{r=1}^T \prod_{\substack{q \mid \Phi_m(mr) \\ q \text{ prime}}} S(q). \quad (\text{BP9})$$

This identity does not assert that either side equals one.

Proof. Use $X^m - 1 = \prod_{d \mid m} \Phi_d(X)$ and $\Phi_m(0) = 1$. For a prime $q \nmid mr$, a root r of Φ_m has order exactly m : a smaller order $e \mid m$ would make it a common root with a factor Φ_d , $d \mid e < m$, and hence a repeated root of $X^m - 1$. This contradicts the nonzero derivative mr^{m-1} modulo q . Conversely an element of order m vanishes on none of the proper-divisor factors, so it vanishes on Φ_m .

For $X = mP$, the constant term gives $\Phi_m(X) \equiv 1 \pmod{X}$, so each prime divisor is prime to mX . Its order is m , which divides $q-1$. Pair primitive roots into nonreal conjugates. For real $X \geq 2$, every pair contributes $|X - \zeta|^2$ strictly between $(X-1)^2$ and $(X+1)^2$; hence

$$(X-1)^{\varphi(m)} < \Phi_m(X) < (X+1)^{\varphi(m)}.$$

This proves $\Phi_m(mP) > 1$. If P is the product of any finite list of primes 1 modulo m , a prime divisor of that value is outside the list and in the same class, proving infinitude.

For a prescribed prime $p \equiv 1 \pmod{m}$, the cyclic group $\mathbb{F}_p^\times$ has exactly $\varphi(m)$ elements of order m . Multiplication by the unit m^{-1} gives exactly that many roots of $\Phi_m(mX)$. As $4 \mid m$, negation preserves primitive m -th roots: the exponent of a primitive root changes from a unit

$k \pmod{m}$ to $k + m/2$, still a unit at every odd prime and still odd. Thus Φ_m is even. The nonzero representatives pair as $r, p - r$, giving the exact half-interval count and in fact

$$p^{\varphi(m)/2} \mid \prod_{r=1}^{(p-1)/2} \Phi_m(mr).$$

This proves (BP8). Every prime occurring in the right side of (BP9) is in the required class, and (BP8) eventually includes every member of that class. Repetitions have no effect because $S(q)^e = S(q)$ for $e \geq 1$. Both infima therefore vanish for exactly the same zero factors and otherwise equal one. $\square$

The polynomial change needed for the larger modulus is exact:

$$\Phi_{12}(X^n) = \prod_{\substack{d \mid n \\ \gcd(n/d, 6)=1}} \Phi_{12d}(X), \quad \Phi_{12n}(X) = \prod_{\substack{d \mid n \\ \gcd(d, 6)=1}} (X^{4n/d} - X^{2n/d} + 1)^{\mu(d)}. \quad (\text{BP10})$$

Indeed $\text{ord}(\zeta^n) = \text{ord}(\zeta) / \gcd(\text{ord}(\zeta), n) = 12$ precisely when $\text{ord}(\zeta) = 12d$, $d \mid n$, and $\gcd(n/d, 6) = 1$. Simple roots give the first polynomial identity; multiplicative Möbius inversion gives the second. Thus $\Phi_{12n}(X) = X^{4n} - X^{2n} + 1$ exactly when $n = 2^\alpha 3^\beta$. Any other prime divisor of n gives an additional nonconstant factor in the first identity. Changing only the argument does not impose order $12n$: $\Phi_{12}(24) = 13 \cdot 73 \cdot 349$.

The recursion $P_0 = 1$, $W_k = \Phi_m(mP_k)$, $P_{k+1} = P_k W_k$ has $P_k = \prod_{i < k} W_i$ and pairwise coprime $W_i > 1$, by the same constant-term calculation. It therefore introduces a new prime at every step. Multiplying an input also by a factorial $b!$ excludes every prime at most b . These statements prove newness and, with such a factorial, strict growth; (BP8)–(BP9) concern the entire family and do not assert coverage by a particular single recursion. These are the mathematical scopes of transcript lines 5438–6411 and 8031–8136.

Theorem 2.8 (Short homogeneous Φ_{12} factorization). *For every prime $p \equiv 1 \pmod{12}$, there exist coprime positive integers $r, s \leq \lfloor \sqrt{p} \rfloor$ such that*

$$r^4 - r^2 s^2 + s^4 = ph, \quad 1 \leq h < p. \quad (\text{BP11})$$

Every prime divisor of h is congruent to 1 modulo 12.

Proof. Choose w of order 12 modulo p and put $H = \lfloor \sqrt{p} \rfloor$. The map $(i, j) \mapsto i - wj$ from $\{0, \dots, H\}^2$ into $\mathbb{F}_p$ has a repeated value because $(H + 1)^2 > p$. The difference gives $(r_0, s_0) \neq (0, 0)$, $|r_0|, |s_0| \leq H$, $r_0 \equiv ws_0 \pmod{p}$. Neither coordinate vanishes: one zero would force the other to be a multiple of p of absolute value less than p . Retain $g = \gcd(|r_0|, |s_0|)$ and their two signs, and set $r = |r_0|/g$, $s = |s_0|/g$. These retained data recover the original signed pair. Since $g < p$, division by it is valid modulo p . Evenness in each coordinate gives $p \mid F(r, s) := r^4 - r^2 s^2 + s^4$.

One has $F = (r^2 - s^2)^2 + r^2 s^2 > 0$. If $r \geq s$, then $F = r^4 - s^2(r^2 - s^2) \leq r^4$, and the opposite case is symmetric. Thus $0 < F \leq H^4 < p^2$, giving (BP11). If a prime $q \mid F$ divided r or s , it would divide both, contrary to coprimality. A primitive pair gives $F \equiv 1 \pmod{2}$ and $F \equiv 1 \pmod{3}$, so these two primes are excluded. The unit rs^{-1} is a root of Φ_{12} modulo q , and the order argument above gives $q \equiv 1 \pmod{12}$. $\square$

This proves the assistant proposition at transcript lines 6612–6751 with the terminal alternative $h = 1$ retained. For comparison of the cofactor and quadratic sign, there is an exact lift formula. If $F(r, s) = ph$, $p \nmid rs$, then

$$\frac{F(r + pt, s)}{p} \equiv h + t 2r(2r^2 - s^2) \pmod{p}. \quad (\text{BP12})$$

The coefficient is a unit: $2r^2 = s^2$ would imply $F = 3s^4/4 = 0 \pmod{p}$, impossible. Hence unrestricted lifts realize every quotient residue modulo p , including either nonzero square class. Primitivity can be retained by choosing t to avoid the one forbidden class modulo each prime dividing s , using CRT together with the prescribed class modulo p . A positive sufficiently large representative gives a positive lift. This exact statement does not retain the short-box bound.

2.4 Necessary nonresidue factors and the exact character target

Lemma 2.9 (Divisor-of-a-sum obstruction). *Let $a, b, d > 0$, $\gcd(a, b) = 1$, $d \mid a + b$, and $d \equiv 3 \pmod{4}$. Then $-d$ is not a square modulo $4ab$.*

Proof. Suppose $t^2 \equiv -d \pmod{4ab}$. The hypotheses give $\gcd(d, ab) = 1$. For each odd prime $\ell \mid ab$, $\left(\frac{-d}{\ell}\right) = 1$. Quadratic reciprocity, including its sign because $d \equiv 3 \pmod{4}$, gives

$$\left(\frac{\ell}{d}\right) = \left(\frac{-d}{\ell}\right) = 1.$$

If $2 \mid ab$, reducing modulo 8 makes t odd and gives $d \equiv 7 \pmod{8}$, so $\left(\frac{2}{d}\right) = 1$. Prime factorization and multiplicativity therefore imply $\left(\frac{a}{d}\right) = \left(\frac{b}{d}\right) = 1$. But $a \equiv -b \pmod{d}$ gives $\left(\frac{a}{d}\right) = -\left(\frac{b}{d}\right)$, a contradiction. $\square$

Theorem 2.10 (A nonresidue prime in every normalized witness). *Every middle witness in (BP1) has a prime divisor $\ell \mid AB$ with $\left(\frac{\ell}{p}\right) = -1$. Every exterior witness has such a prime divisor of BC .*

Proof. For the middle channel, $R \mid A+B$, $\gcd(A, B) = 1$, and $p \equiv -R \pmod{4AB}$. Lemma 2.9 says p is not a square modulo $4AB$. If every prime $\ell \mid AB$ were a residue modulo p , reciprocity would give $\left(\frac{p}{\ell}\right) = 1$ for each odd such prime, because $p \equiv 1 \pmod{4}$. A unit square root modulo ℓ lifts to every power: given $x^2 \equiv p \pmod{\ell^e}$, solve $2xt \equiv (p - x^2)/\ell^e \pmod{\ell}$ and replace x by $x + t\ell^e$. If $2 \mid AB$, $\left(\frac{2}{p}\right) = 1$ and $p \equiv 1 \pmod{4}$ give $p \equiv 1 \pmod{8}$. Every such unit is a square modulo each 2-power: starting modulo 8, replacing an odd root x modulo 2^e , $e \geq 3$, by $x + t2^{e-1}$ toggles the next square bit modulo 2^{e+1} . If AB is odd, the needed modulo-4 root follows from $p \equiv 1 \pmod{4}$. CRT now makes p a square modulo $4AB$, a contradiction.

For the exterior channel the tuple equation is

$$A(4BCD - 1) = p(B + C).$$

Since $A < p$, set $T = (B + C)/A \in \mathbb{Z}_{>0}$. Then $pT + 1 = 4BCD$, $T \equiv 3 \pmod{4}$, and $T \mid B + C$. A common prime divisor of B, C would divide A , so $\gcd(B, C) = 1$. Also $p \nmid BC$: $B < p$, and $p \mid C$ in the tuple equation would force $p \mid A$. If p were a square modulo $4BC$, its inverse would be one, contrary to Lemma 2.9 because $p^{-1} \equiv -T \pmod{4BC}$. The same prime-power and reciprocity argument therefore supplies a nonresidue prime in BC . $\square$

For any fixed $K \geq 2$, put

$$m_K = \text{lcm} \left(12n, 8, \prod_{\substack{3 \leq \ell \leq K \\ \ell \text{ prime}}} \ell \right).$$

Every prime $p \equiv 1 \pmod{m_K}$ makes all primes at most K quadratic residues: use reciprocity for odd primes and the modulo-8 supplementary law for 2. Theorem 2.10 therefore forces a

prime greater than K in the specified product AB or BC . In the middle channel, $A \leq 2$ forces that prime to divide B , hence $B > K$. Theorem 2.7 produces infinitely many primes satisfying these restrictions. This excludes fixed bounds on these pairs, not all witnesses or the global two-layer assertion. These are the exact quantifiers of transcript lines 6836–7265.

Proposition 2.11 (A nonempty Jacobi-sign class and exact target counts). *Let $p \geq 13$ be prime and $p \equiv 1 \pmod{4}$, and put $L = (p-1)/4$. There is a prime quadratic nonresidue $\ell \leq L$. The integer $a = \ell \lceil (L+1)/\ell \rceil$ is a first-half shell denominator containing a nonresidue prime factor. For this shell put*

$$\tau = \prod_{r|a} (2v_r(a) + 1), \quad \sigma = \prod_{\substack{r|a \\ (r/p)=1}} (2v_r(a) + 1).$$

Then the exact number of divisors $u \mid a^2$ with $(\frac{u}{R}) = -1$ is $(\tau - \sigma)/2 \geq \tau/3 > 0$.

For every first-half shell, independently of this choice, let

$$W_a = \left\{ \prod_{\ell|a} \ell^{e_\ell} \pmod{R} : -v_\ell(a) \leq e_\ell \leq v_\ell(a) \right\}.$$

Then $W_a^{-1} = W_a$ and

$$E_a \cup M_a \neq \emptyset \iff \{-1, -p\} \cap W_a \neq \emptyset. \quad (\text{BP13})$$

With ψ ranging over all characters of $(\mathbb{Z}/R\mathbb{Z})^\times$,

$$|E_a| + |M_a| = \frac{1}{\varphi(R)} \sum_{\psi} \psi(-1)(1 + \psi(p)) \prod_{\ell^v \parallel a} \sum_{e=-v}^v \psi(\ell)^e. \quad (\text{BP14})$$

Proof. If every integer 1 through L were a residue, their negatives would also be residues because $p \equiv 1 \pmod{4}$, exhausting all $2L$ nonzero residues. Every integer from $L+1$ through $3L$ would then be a nonresidue. But $2L = 2 \cdot L$ is a product of two assumed residues and lies in that interval. The least nonresidue is prime because otherwise all its prime factors are smaller residues. The prescribed a satisfies $L+1 \leq a \leq L+\ell \leq 2L$.

For an odd prime $r \mid a$, reciprocity and $R \equiv -p \pmod{r}$, $R \equiv 3 \pmod{4}$, give $(\frac{r}{R}) = (\frac{r}{p})$: the two reciprocity signs cancel. For $r = 2$, evenness of a gives $R \equiv -p \pmod{8}$ and the same identity by the supplementary law. Summing the Jacobi symbols of divisors of a^2 therefore factors into geometric sums. A residue prime contributes $2v+1$, and a nonresidue prime contributes $\sum_{e=0}^{2v} (-1)^e = 1$. The sum is σ , the total is τ , and the negative count follows. At least one nonresidue factor gives $\tau \geq 3\sigma$.

Dividing $u \mid a^2$ by a translates exponents by $-v_\ell(a)$, and inversion negates every centered exponent. The middle target becomes -1 , while the exterior target becomes $-p^{-1}$; inversion changes it to $-p$. This proves (BP13). Orthogonality gives, for a unit t ,

$$\#\{u \mid a^2 : u \equiv t \pmod{R}\} = \frac{1}{\varphi(R)} \sum_{\psi} \overline{\psi(t)} \prod_{\ell^v \parallel a} \sum_{e=0}^{2v} \psi(\ell)^e.$$

Shift each exponent by $-v$, retaining the factor $\psi(a)$. At $t = -a$ its prefactor is $\psi(-1)$; at $t = -4^{-1}$ it is $\psi(-1)\psi(4a) = \psi(-1)\psi(p)$. Adding gives (BP14). No individual summand is asserted nonnegative; the total is an integer target count. $\square$

There is a further exact comparison with the divisor-pair statistic used in the dialogue. Let

$$T_{R,S} = \#\{(m, r) : m, r \mid S, R \mid m + r\}.$$

Character orthogonality gives

$$T_{R,S} = \frac{1}{\varphi(R)} \sum_{\psi} \psi(-1) \left| \sum_{d \mid S} \psi(d) \right|^2. \quad (\text{BP15})$$

Indeed expansion has the target $-mr^{-1} = 1$ in the unit group. Reduction by $g = \gcd(m, r)$ maps these pairs onto the primitive pairs (m_0, r_0) with $m_0, r_0 \mid S$, $\gcd(m_0, r_0) = 1$, $R \mid m_0 + r_0$. Its full fibre is

$$\{(gm_0, gr_0) : g \mid S/(m_0r_0)\}. \quad (\text{BP16})$$

To prove this, division by $g \mid S$ preserves the congruence because g is a unit modulo R . Conversely both scaled entries divide S exactly when $g \mid \gcd(S/m_0, S/r_0) = S/(m_0r_0)$. The primitive pair reconstructs the ordered solution

$$k = (m_0 + r_0)/R, \quad (a, y, z) = (a, (S/r_0)k, (S/m_0)k).$$

Its reciprocal sum is $1/a + R/S = 4/p$. The inverse primitive pair is the reduced positive ratio $(Ry - S)/S = y/z$; prime valuations in $(Ry - S)(Rz - S) = S^2$ prove both its entries divide S . Thus $T > 0$ is exactly shell occupancy, while its multiplicities are the divisor counts of (BP16). A canonical hit supplies the distinct pair $(A, p^{1-\varepsilon}B)$, so $T \geq 2$. This proves, with its full multiplicity map, the relation in transcript lines 3594–3744 and 7843–7887.

2.5 Five exact regressions and their scopes

The following finite certificates are the five stronger-claim counterexamples of transcript lines 8330–8786. They were reproduced by a separate standard-library integer calculation. Their exhaustiveness concerns the displayed finite divisor sets or coordinate boxes, not all primes or all Erdős–Straus candidates.

1. A selected nonresidue shell can be empty. Take $p = 3361$, $L = 840$. The primes 2,3,5,7 are residues modulo p : use $p \equiv 1 \pmod{8}$, $p \equiv 1 \pmod{12}$, and reciprocity with $p \equiv 1 \pmod{5}$, $p \equiv 1 \pmod{7}$. Thus every integer 1 through 10 is a residue. On the other hand $3361 \equiv 6 \pmod{11}$, and the nonzero squares modulo 11 are $\{1, 3, 4, 5, 9\}$, so 11 is the least nonresidue. Proposition 2.11 selects

$$a = 11 \lceil 841/11 \rceil = 847 = 7 \cdot 11^2, \quad R = 27, \quad j = 6.$$

Every divisor of a^2 is $7^i 11^e$, $0 \leq i \leq 2$, $0 \leq e \leq 4$. Their residues are exactly

$$\{1, 2, 7, 8, 10, 11, 13, 14, 16, 19, 22, 23, 26\} \pmod{27}.$$

They omit $-4^{-1} = 20$ and $-a = 17$, proving $E_6 = M_6 = \emptyset$. The six negative-Jacobi divisors are 11, 77, 539, 1331, 9317, 65219. The principal contribution to (BP14) is $2 \cdot 15/18 = 5/3$, while its complete sum is zero; the other characters contribute $-5/3$. This is not an ES counterexample: the first shell has $a = 841$, $R = 3$, $u = 29 \mid 841^2$, and $3 \mid 4 \cdot 29 + 1$. Its tuple is $(A, B, C, D, \varepsilon) = (29, 1, 1130, 29, 0)$, giving

$$c_H = 79027200, \quad \frac{4}{3361} = \frac{1}{841} + \frac{1}{950330} + \frac{1}{110139970}.$$

2. Nonempty local fibres need not intersect in the divisor box. At $p = 37, a = 18, R = 35$, the complete divisor list is

$$\text{Div}(324) = \{1, 2, 3, 4, 6, 9, 12, 18, 27, 36, 54, 81, 108, 162, 324\}.$$

Those satisfying $5 \mid 4u + 1$ are 1, 6, 36, 81; those satisfying $7 \mid 4u + 1$ are 12, 54. Both lists are nonempty and their intersection is empty. CRT gives $u \equiv 26 \pmod{35}$, but no member of the original complete list belongs to that class. This refutes the exchange of $\forall m \exists u$ with $\exists u \forall m$, while preserving Theorem 2.6.

3. The short cofactor may always be one at a nonbase prime. The complete positive primitive boxes in Theorem 2.8 give the following table:

p	H	all primitive pairs	h
61	7	(2, 3), (3, 2)	1
73	8	(1, 3), (3, 1)	1
193	13	(3, 4), (4, 3)	1
2521	50	(1, 26), (26, 1)	181

Here is a finite exact verification of completeness. The roots of Φ_{12} in these four prime fields are, respectively,

$$\{21, 29, 32, 40\}, \quad \{3, 24, 49, 70\}, \quad \{49, 63, 130, 144\}, \quad \{26, 97, 2424, 2495\}.$$

Substitution gives four distinct roots in each case, so the polynomial degree proves the lists complete. For each root w , the possible positive coordinates in the box are exactly

$$r = ws - pk, \quad \max\left(1, \left\lceil \frac{pk + 1}{w} \right\rceil\right) \leq s \leq \min\left(H, \left\lfloor \frac{pk + H}{w} \right\rfloor\right), \quad 0 \leq k \leq \lfloor wH/p \rfloor. \quad (\text{BP17})$$

Evaluating these integer intervals gives, before removing nonprimitive pairs, respectively

61	(2, 3), (3, 2), (4, 6), (6, 4)
73	(1, 3), (2, 6), (3, 1), (6, 2)
193	(3, 4), (4, 3), (6, 8), (8, 6), (9, 12), (12, 9)
2521	(1, 26), (26, 1).

The gcd test and direct quartic evaluation give the preceding table. For example $2^4 - 2^2 3^2 + 3^4 = 61$. Hence $h > 1$ cannot be demanded at every prime greater than 13 in this prescribed box.

4. A smaller cyclotomic prime need not be a nonresidue. The last row of that complete table gives

$$1 - 26^2 + 26^4 = 456301 = 2521 \cdot 181, \quad 88^2 = 3 \cdot 2521 + 181.$$

Thus $181 < 2521$, $181 \equiv 1 \pmod{12}$, but $\left(\frac{181}{2521}\right) = 1$. Both permitted pairs give that cofactor. The actual middle tuple $(A, B, C, D, \varepsilon) = (2, 159, 7, 2, 1)$ has $AB = 2 \cdot 3 \cdot 53$, with 53 a nonresidue modulo 2521: reciprocity gives

$$\left(\frac{53}{2521}\right) = \left(\frac{30}{53}\right) = \left(\frac{2}{53}\right) \left(\frac{3}{53}\right) \left(\frac{5}{53}\right) = (-1)(-1)(-1) = -1.$$

Its code and original denominators are

$$c_H = 1600517, \quad \frac{4}{2521} = \frac{1}{636} + \frac{1}{70588} + \frac{1}{5611746}.$$

Substitution gives $4ABCD = 17808 = A + B + pC$, and all the code bits equal one. Thus the cofactor and witness factor have their respective exact roles.

5. The degree-four cofactor bound does not persist for Φ_{24} . For $p = 73$, the eight roots of $\Phi_{24}(X) = X^8 - X^4 + 1$ modulo 73 are 7, 17, 21, 30, 43, 52, 56, 66. Substitution and degree prove completeness. Formula (BP17), now with $H = 8$ and this root list, gives exactly $(1, 7), (7, 1), (4, 5), (5, 4)$. They are all primitive, and

$$F_{24}(1, 7) = 5762401 = 73 \cdot 78937, \quad F_{24}(4, 5) = 296161 = 73 \cdot 4057.$$

The smallest quotient is $4057 > 73$, disproving the unchanged requirement $h < p$. The valid bound for this degree-eight trinomial is $F_{24}(r, s) \leq \max(r, s)^8 < p^4$, hence $h < p^3$. The cyclotomic infinitude theorem at modulus 24 is unaffected.

The primality checks needed in these finite certificates are elementary trial divisions by all primes at most the respective square roots. For 61 and 73 those primes are 2, 3, 5, 7; for 193 they are 2, 3, 5, 7, 11, 13; for 2521 they are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47; for 3361 add 53 to that last list. None divides the corresponding number. For the auxiliary primes 97, 181, 197 the same test uses respectively the lists through 7, 13, 13. The complete finite divisor sets, root lists, quotient values and reciprocal identities above are exact certificates for their stated counterexamples. They do not establish the universal positivity of (BP3) or (BP9).

3 Exact character-resolved collision minima and integral ES return

Provenance and scope. This module reconstructs the character-resolved energy result stated in the supplied joint ES discussion, `pasted-text.txt`, in its final “Today 8:19 AM” entry, subsection “JT’s contribution supplies an exact refinement of the positivity test.” The discussion credits the quadratic character refinement to its integration of JT’s bounded-divisor synthesis. The original `src/energy.tex` was not available: the downloaded loose `workbench.tex` is a wrapper referring to that missing file. The shell and ordered channel conventions are those in the supplied earlier `es_dual_frontier/workbench.tex` and the definition block in `dual_es.py`; their required identities are proved here in full. The independently authored checker imports no downloaded code. The result is a sufficient test on a specified full exponent box. It proves no universal statement that every prime has a shell passing this test.

3.1 The complete shell, exponent box, and ordered return

Fix a prime $p = 4m + 1$, where $m \geq 1$, and an integer

$$m + 1 \leq a \leq 3m, \quad R = 4a - p.$$

Then $3 \leq R \leq 8m - 1 = 2p - 3$, $R \equiv 3 \pmod{4}$, and $0 < a < p$. Since p is prime, $\gcd(a, p) = 1$; consequently

$$\gcd(a, R) = \gcd(a, p) = 1, \quad \gcd(p, R) = \gcd(p, 4a) = 1.$$

Write the complete prime factorization as $a = \prod_{q|a} q^{e_q}$. All these primes are units modulo R . Define

$$B_a = \prod_{q|a} ([-e_q, e_q] \cap \mathbb{Z}), \quad v(\beta) = \prod_{q|a} q^{\beta_q} \in G = (\mathbb{Z}/R\mathbb{Z})^\times, \quad u(\beta) = \prod_{q|a} q^{e_q + \beta_q}.$$

Negative powers in v mean multiplicative inverses in G . Unique prime factorization proves that $\beta \mapsto u(\beta)$ is a bijection from B_a onto the positive divisors of a^2 : its inverse is $\beta_q = v_q(u) - e_q$. Multiplying the residue products gives

$$v(\beta) = u(\beta)a^{-1} \pmod{R}, \quad u(-\beta) = a^2/u(\beta), \quad v(-\beta) = v(\beta)^{-1}.$$

In particular no unbounded subgroup or shorter exponent interval has replaced B_a .

For any positive divisor $u \mid a^2$, put $d = a^2/u$ and retain the two ordered rational triples

$$Q_E(p, a, u) = \left(a, \frac{pa + d}{R}, \frac{pa + p^2u}{R} \right),$$

$$Q_M(p, a, u) = \left(a, \frac{p(a + d)}{R}, \frac{p(a + u)}{R} \right).$$

Their first denominator remains distinguished; the last two entries are not sorted. In either case, if these entries are denoted Y, Z , then

$$(RY - pa)(RZ - pa) = p^2a^2.$$

Indeed the two factors are (d, p^2u) in channel E and (pd, pu) in channel M . Expanding and cancelling the identical constant term gives $R^2YZ - paR(Y + Z) = 0$. Since R, p, a, Y, Z are positive, division gives

$$\frac{1}{Y} + \frac{1}{Z} = \frac{R}{pa}, \quad \frac{1}{a} + \frac{1}{Y} + \frac{1}{Z} = \frac{p + R}{pa} = \frac{4}{p}.$$

Both triples therefore always give positive rational ES decompositions.

The exact integral gates follow without discarding the modulus R . For E , multiplication of the first numerator by the unit u yields

$$u(pa + d) \equiv a^2(4u + 1) \pmod{R}, \quad pa + p^2u \equiv 4a^2(4u + 1) \pmod{R}.$$

For M the exact numerator identities are

$$up(a + d) = pa(u + a), \quad p(a + u) = p(u + a).$$

Since every multiplier used is a unit modulo R , the two reduced denominators in each channel coincide and are respectively

$$D_E = \frac{R}{\gcd(R, 4u + 1)}, \quad D_M = \frac{R}{\gcd(R, u + a)}.$$

Thus Q_E is integral exactly when $R \mid 4u + 1$, and Q_M exactly when $R \mid u + a$. This proves all assertions about integrality, not merely one implication.

Since $u = av(\beta)$ modulo R and $4a = p$ modulo R , these tests become

$$M : v(\beta) = -1, \quad E : v(\beta) = -p^{-1}.$$

The complete box is stable under $\beta \mapsto -\beta$, and $(-p)^{-1} = -p^{-1}$, so a vector with residue $-p$ also supplies an exterior witness by applying this involution. Consequently simultaneous failure of both channels in this shell is equivalent to absence of all the residues in the deduplicated, inversion-stable set

$$T = \{-1, -p, -p^{-1}\} \subseteq G.$$

The set may have fewer than three elements, and may contain the identity; both possibilities are retained below.

For the additional integral marking put

$$g = \gcd(a, u), \quad h = g^2/u, \quad r = u/g, \quad s = a/g.$$

If $e = v_q(a)$ and $f = v_q(u)$, then $0 \leq f \leq 2e$ and the exponents of h, r, s are $2 \min(e, f) - f$, $f - \min(e, f)$ and $e - \min(e, f)$. The first is f when $f \leq e$ and $2e - f$ otherwise, so all are

nonnegative. At least one of the last two is zero, proving $\gcd(r, s) = 1$. Direct substitution now proves

$$a = hrs, \quad u = hr^2, \quad d = hs^2,$$

and the ordered triples are

$$Q_E = (hrs, hsk, phrk), \quad \kappa = (pr + s)/R, \quad Q_M = (hrs, phs\lambda, phr\lambda), \quad \lambda = (r + s)/R.$$

Since g is a unit modulo R , the middle gate is equivalent to $R \mid r + s$. The exterior numerator satisfies $pa + d = hs(pr + s)$, where hs is a unit modulo R , so the exterior gate is equivalent to $R \mid pr + s$. Hence these quotients are positive integers whenever the corresponding target occurs. Every marking returns to the original objects via $a = hrs$, $u = hr^2$ and $\beta_q = v_q(u) - e_q$; moreover

$$u = \frac{a^2}{RY - pa} \quad (E), \quad u = \frac{pa^2}{RY - pa} \quad (M).$$

Thus the channel and ordered denominators retain an explicit inverse.

For completeness, the full interval of shells suffices for the ES existence question at this p . Given any positive integral solution, move a smallest denominator to the distinguished position a , retaining that permutation. The inequalities $4/p > 1/a$ and $4/p \leq 3/a$ give $p/4 < a \leq 3p/4$, hence $m + 1 \leq a \leq 3m$. Its other two denominators obey the residual product identity above, and $RY - pa = paY/Z > 0$, $RZ - pa = paZ/Y > 0$. Since $p \nmid a$, the first residual is uniquely $p^i v$ with $i \in \{0, 1, 2\}$ and $v \mid a^2$. For $i = 0$, take $E, u = a^2/v$ in the existing order. For $i = 1$, take $M, u = a^2/v$ in the existing order. For $i = 2$, take $E, u = v$ and retain the swap of the last two entries. The displayed residuals verify each case. Conversely, an integral candidate already solves ES. This proves the exact equivalence with a hit in the full atlas; it does not assert that such a hit always exists.

3.2 Multiplicities, parity, and character cells

Define

$$\mu(g) = \#\{\beta \in B_a : v(\beta) = g\}, \quad N = \sum_{g \in G} \mu(g) = \prod_{q \mid a} (2e_q + 1), \quad E = \sum_{g \in G} \mu(g)^2.$$

The involution $\beta \mapsto -\beta$ bijects the fibres of g and g^{-1} , proving $\mu(g) = \mu(g^{-1})$. Its only fixed vector in the integer box is zero. The fibre over 1 contains that vector and otherwise splits into two-element orbits; thus $\mu(1)$ is odd and at least one. If $g^2 = 1$ but $g \neq 1$, its fibre is stable under this involution and contains no fixed vector, so $\mu(g)$ is even.

The collision count also has the independent exact expression

$$E = \sum_{\substack{-2e_q \leq \delta_q \leq 2e_q \\ \prod q^{\delta_q} = 1 \pmod{R}}} \prod_{q \mid a} (2e_q + 1 - |\delta_q|).$$

Indeed E counts ordered pairs (β, γ) with identical residue. Their coordinate difference $\delta = \beta - \gamma$ has residue one. For a single interval $[-e, e]$, the number of ordered pairs of difference δ is $2e + 1 - |\delta|$ for $|\delta| \leq 2e$, and zero otherwise: the first entry lies in $[-e, e] \cap [-e + \delta, e + \delta]$. Taking the product of these counts proves the formula.

Let V be a finite elementary abelian 2-group, written multiplicatively, and let $\pi : G \rightarrow V$ be a homomorphism; surjectivity is not required. For $v \in V$ put

$$C_v = \pi^{-1}(v), \quad N_v = \sum_{g \in C_v} \mu(g), \quad E_v = \sum_{g \in C_v} \mu(g)^2.$$

As $v^{-1} = v$, each cell is stable under inversion. Its mass is even except in the cell containing 1, where its mass is odd: inverse pairs contribute twice their common multiplicity, nonidentity involutions have even multiplicity, and the identity contributes odd multiplicity. Empty cells have mass and energy zero.

For each odd prime $\ell \mid R$, the prime-local quadratic character $\chi_\ell(g)$ is 1 when g is a nonzero square modulo ℓ and -1 otherwise. This is a homomorphism: in the multiplicative group of the field $\mathbb{Z}/\ell\mathbb{Z}$, the square map is a homomorphism whose kernel is exactly $\{1, -1\}$, since $(x-1)(x+1) = 0$ implies $x = 1$ or $x = -1$ in a field. Every fibre has two elements, so the square subgroup has index two. The quotient by that subgroup has two elements and therefore supplies precisely this sign character. Reduction modulo ℓ is a group homomorphism on G , which proves the assertion for χ_ℓ on G as well.

Writing $R = \prod_{\ell \mid R} \ell^{f_\ell}$, define

$$\pi_{\text{quad}}(g) = (\chi_\ell(g))_{\ell \mid R}, \quad \pi_{\text{Jac}}(g) = \prod_{\ell \mid R} \chi_\ell(g)^{f_\ell}.$$

Both are homomorphisms. The second is obtained from the first by the explicit homomorphism $(\epsilon_\ell) \mapsto \prod \epsilon_\ell^{f_\ell}$; factors with even f_ℓ disappear only from this product. For example, modulo 9 the residue 2 is a non-square modulo 3, so its prime-local character is -1 , whereas its Jacobi character is $(-1)^2 = 1$. Thus these two partitions need not coincide. The trivial homomorphism is the unpartitioned case.

3.3 The exact minimum, including all impossible cases

For a fixed cell let $A_v = C_v \setminus T$. Since both C_v and T are inversion-stable, so is A_v . Its slots are the unordered inverse pairs $\{g, g^{-1}\}$ with $g \neq g^{-1}$, the individual nonidentity involutions $g^2 = 1$, $g \neq 1$, and the identity if it is present. Write $b_v = 1$ when $1 \in C_v$ and $b_v = 0$ otherwise.

For a nonnegative integer mass n , define $L_v(n)$ as the minimum of $\sum_{g \in C_v} \nu(g)^2$ over all maps $\nu : C_v \rightarrow \mathbb{Z}_{\geq 0}$ satisfying

$$\sum_{g \in C_v} \nu(g) = n, \quad \nu(g) = \nu(g^{-1}), \quad \nu|_{T \cap C_v} = 0,$$

and additionally $\nu(1)$ odd when $1 \in C_v$, and $\nu(g)$ even for every nonidentity involution in the cell. If there is no such map, set $L_v(n) = +\infty$.

Here is a complete evaluation. If $1 \in T \cap C_v$, the minimum is $+\infty$ for every n , since an odd nonnegative identity value cannot vanish. Otherwise the identity slot exists precisely when $b_v = 1$. If $n < b_v$ or $n - b_v$ is odd, the minimum is $+\infty$. In all remaining cases put $K = (n - b_v)/2$. Write P_v for the set of inverse-pair slots and J_v for the nonidentity involution slots. Use nonnegative integer variables k_P for $P \in P_v$, z_j for $j \in J_v$, and k_0 only if $b_v = 1$. Then

$$L_v(n) = \min_{\sum_P k_P + \sum_j z_j + b_v k_0 = K} \left(2 \sum_P k_P^2 + 4 \sum_j z_j^2 + \begin{cases} (1 + 2k_0)^2, & b_v = 1, \\ 0, & b_v = 0. \end{cases} \right),$$

where the absent variable is omitted from both sums. If $K > 0$ and there are no slots, the minimum is $+\infty$; if $K = 0$ it is b_v . These are all the impossible cases: when at least one slot exists, it can hold every remaining pair of mass units. For a pair slot the two values are (k_P, k_P) , for a nonidentity involution its value is $2z_j$, and for the identity its value is $1 + 2k_0$. These assignments give a bijection between feasible slot allocations and feasible maps ν , proving the optimization formula. There are finitely many allocations for fixed K , so every finite minimum is attained.

Start the identity at mass one and energy one when $b_v = 1$ and start every other slot at zero. Incrementing a slot with current index k by one adds two units of mass. Its exact energy increment is respectively

slot	current values	incremental energy
$\{g, g^{-1}\}, g \neq g^{-1}$	(k, k)	$2(k+1)^2 - 2k^2 = 4k + 2$
$g^2 = 1, g \neq 1$	$2k$	$(2k+2)^2 - (2k)^2 = 8k + 4$
$g = 1$	$1 + 2k$	$(3+2k)^2 - (1+2k)^2 = 8k + 8$

In each row k starts at zero. Thus the marginal sequences are $2, 6, 10, \dots$, $4, 12, 20, \dots$, and $8, 16, 24, \dots$. Selecting the K least costs from the union of these increasing sequences proves the minimum exactly. To justify feasibility, every predecessor in a sequence is strictly smaller than its successor, so selecting a successor among the K least costs selects its whole prefix. Ties between different sequences can be resolved in any fixed order. To justify optimality, any feasible allocation selects exactly K costs from this union, hence its sum is at least the sum of the K least ones. Equivalently, replacing a selected unnecessarily large cost by the least unselected available cost improves the allocation until the same minimum is reached. A heap holding the first unused cost of each sequence implements this finite procedure.

Forcing theorem. For the actual full-box multiplicities and every cell v ,

$$\mu|_{T \cap C_v} = 0 \implies E_v \geq L_v(N_v).$$

Indeed all other constraints defining L_v were proved for μ , so this restriction is a feasible map if its target fibres vanish. Taking the contrapositive proves

$$E_v < L_v(N_v) \implies \text{there is } \beta \in B_a \text{ with } v(\beta) \in T \cap C_v.$$

The preceding explicit map then produces a positive integral ES witness. In particular, if $T \cap C_v = \{-1\}$, the witness is in the middle channel. A finite energy is smaller than $+\infty$, so incompatible cells are included without an extra convention. If $1 \in T$, the zero vector itself is already a target and is detected by this rule. If $T \cap C_v$ is empty, the actual restriction is feasible and this cell can never produce a false certificate.

Every finite $L_v(n)$ is attained by a distribution satisfying exactly the specified mass, symmetry, parity, and exclusions. Hence it is the largest universal lower bound on energy from those constraints alone. This is the precise optimality assertion. It does not state that every energy above the minimum is attainable, or that every attaining abstract distribution arises from the arithmetic exponent box.

3.4 Exact refinement and preservation of certificates

Let $\pi_2 : G \rightarrow V_2$ and $\pi_1 : G \rightarrow V_1$ be such homomorphisms, and suppose $\pi_1 = \rho \circ \pi_2$ for a homomorphism $\rho : V_2 \rightarrow V_1$. A coarse cell C_v is the disjoint union of the fine cells C_w with $\rho(w) = v$. For any prescribed fine masses n_w , with $n = \sum_{\rho(w)=v} n_w$, the exact inequality is

$$L_v^{(1)}(n) \leq \sum_{\rho(w)=v} L_w^{(2)}(n_w).$$

All terms take values in $\mathbb{Z}_{\geq 0} \cup \{+\infty\}$. If some right-hand term is infinite the inequality is automatic. Otherwise choose a minimizer on each fine cell and join these maps. Each inversion orbit lies entirely in a fine cell, so the joined map obeys every coarse constraint, with mass n

and energy equal to the sum on the right. This proves the inequality. Restricting any coarse feasible map to its fine cells proves the stronger exact identity

$$L_v^{(1)}(n) = \min_{\substack{n_w \in \mathbb{Z}_{\geq 0} \\ \sum_{\rho(w)=v} n_w = n}} \sum_{\rho(w)=v} L_w^{(2)}(n_w).$$

If the coarse minimum is finite, its minimizer supplies the masses and gives the reverse inequality. If it is infinite, no combination of finite fine minimizers can exist, since their join would be a coarse feasible map. This also proves equality in the infinite case.

For actual masses and energies, $N_v^{(1)} = \sum N_w^{(2)}$ and $E_v^{(1)} = \sum E_w^{(2)}$ by disjointness. If every fine cell satisfies $E_w^{(2)} \geq L_w^{(2)}(N_w^{(2)})$, their sum and the inequality above give $E_v^{(1)} \geq L_v^{(1)}(N_v^{(1)})$. The contrapositive proves that every coarse forcing certificate yields a forcing certificate in at least one of its fine cells. This applies successively to the trivial, Jacobi, and all-prime-local quadratic partitions, with the explicit product map already proved above.

3.5 The complete numerical witness at $p = 1201$

Take $p = 1201$, $a = 310 = 2 \cdot 5 \cdot 31$, $R = 39 = 3 \cdot 13$. To certify primality explicitly, $34^2 < 1201 < 35^2$ and the residues of 1201 modulo the primes 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31 are respectively 1, 1, 1, 4, 2, 5, 11, 4, 5, 12, 23. None is zero, and every composite integer has a prime divisor at most its square root, as follows by writing it as the product of two positive factors larger than one and taking a prime factor of the smaller one. These are all primes at most 34, so 1201 is prime. There are exactly $3^3 = 27$ exponent vectors with each coordinate in $\{-1, 0, 1\}$. In the residue ring, $p = 31$ and $31^{-1} = 34$, since $31 \cdot 34 = 27 \cdot 39 + 1$. Hence $T = \{5, 8, 38\}$. The complete histogram on the 24 units is

g	1	2	4	5	7	8	10	11	14	16	17	19
$\mu(g)$	1	1	1	1	2	1	1	0	2	1	1	2
g	20	22	23	25	28	29	31	32	34	35	37	38
$\mu(g)$	1	1	1	0	2	1	1	0	1	1	2	2

This table can be checked by multiplying all 27 combinations of $(2^{-1}, 1, 2) = (20, 1, 2)$, $(5^{-1}, 1, 5) = (8, 1, 5)$ and $(31^{-1}, 1, 31) = (34, 1, 31)$ modulo 39. Its six entries of value two and fifteen entries of value one give $N = 27$ and $E = 6 \cdot 4 + 15 = 39$.

Deleting T leaves nine inverse pairs,

$$\{2, 20\}, \{4, 10\}, \{7, 28\}, \{11, 32\}, \{16, 22\}, \{17, 23\}, \{19, 37\}, \{29, 35\}, \{31, 34\},$$

the two nonidentity involutions 14, 25, and the identity. The unpartitioned mass budget is $(27 - 1)/2 = 13$. The 13 least marginal costs are nine copies of 2, two copies of 4, and two copies of 6. Thus

$$L_{\text{unpartitioned}}(27) = 1 + 9 \cdot 2 + 2 \cdot 4 + 2 \cdot 6 = 39.$$

Equality $E = 39$ gives no strict forcing certificate at this level.

The negative Jacobi cell is exactly

$$C_- = \{7, 14, 17, 19, 23, 28, 29, 31, 34, 35, 37, 38\}.$$

For a direct sign check the nonzero squares modulo 3 are $\{1\}$, and those modulo 13 are $\{1, 3, 4, 9, 10, 12\}$; taking the product of these two signs gives exactly the displayed cell. Its table entries give $N_- = 18$, $E_- = 30$, and $T \cap C_- = \{38\}$. Removing this target leaves five pairs

$$\{7, 28\}, \{17, 23\}, \{19, 37\}, \{29, 35\}, \{31, 34\}$$

and the nonidentity involution 14. There is no identity slot. The budget is nine, whose nine least costs are five copies of 2, one copy of 4, and three copies of 6. Therefore

$$L_-(18) = 5 \cdot 2 + 4 + 3 \cdot 6 = 32 = \min_{n_1 + \dots + n_5 + z = 9} \left(2 \sum_{i=1}^5 n_i^2 + 4z^2 \right).$$

All variables in the minimum are nonnegative integers. The allocation $(n_1, \dots, n_5, z) = (2, 2, 2, 1, 1, 1)$ attains 32, and the marginal argument proves that no smaller value occurs. Hence $30 < 32$ forces the middle target in this specific cell.

The finer labels are ordered by the primes $(3, 13)$; their complete mass, energy, minimum and target data are

v	N_v	E_v	$L_v(N_v)$	$T \cap C_v$
$(-1, -1)$	4	4	4	$\{5, 8\}$
$(-1, +1)$	8	12	14	$\{38\}$
$(+1, -1)$	10	18	18	$\emptyset$
$(+1, +1)$	5	5	5	$\emptyset$

These values follow by grouping the full histogram using the explicit square lists. The allowed slot counts are respectively two pairs; two pairs and the involution 14; three pairs; and two pairs, the identity, and the involution 25. The marginal formula gives the four displayed minima, including $2 + 2 + 4 + 6 = 14$ for $(-1, +1)$.

An actual middle fibre vector is $\beta = (0, -1, -1)$ in the prime order $(2, 5, 31)$. Its residue is $8 \cdot 34 = 272 = 6 \cdot 39 + 38$, and its original divisor is $u = 2^{1+0}5^{1-1}31^{1-1} = 2$. The complete marking is therefore

$$(p, R, u, h, r, s, \lambda) = (1201, 39, 2, 2, 1, 155, 4),$$

because $\gcd(310, 2) = 2$ and $(1 + 155)/39 = 4$. Substitution in the ordered middle return gives

$$Q_M = (310, 1201 \cdot 2 \cdot 155 \cdot 4, 1201 \cdot 2 \cdot 1 \cdot 4) = (310, 1489240, 9608).$$

Every number is a positive integer. The earlier proved reciprocal identity now verifies the ES equation exactly, and the retained slope satisfies $r/s = 1/155 < 1/8$. This is a stronger forcing certificate for this witness, with its entire divisor vector retained.

3.6 Finite verification and the boundary of the result

The independent standard-library checker compares the marginal algorithm with an independent dynamic program that uses the occupancy squares directly. It includes all masses $0 \leq n \leq 19$, zero through three pair slots, zero through three nonidentity involution slots, presence or absence of the identity, and the forbidden-identity case. It also compares the actual cells of the 1201, 310 example with that dynamic program. The checker uses explicit exceptions, so optimized Python retains every check. For its arithmetic cases it enumerates the full box and independently enumerates divisors of a^2 by trial divisibility and complementary divisors. The signed-box symmetries, parity, target exclusions, refinement inequalities, and extracted rational identities are checked exactly. The weighted autocorrelation is independently checked on the listed examples, including shells with $R \geq p$.

The scan uses exactly the primes $p \equiv 1 \pmod{12}$ with $p \leq 1500$ and the first-half shells $m + 1 \leq a \leq 2m$, equivalently $R < p$. It contains 54 primes and 9531 shells and gives respectively 214 unpartitioned, 231 Jacobi, and 240 fully quadratic certifications. These are shell counts. The general theorem retains the full shell range through $3m$; this finite scan does not replace that range. The empty shell $p = 37, a = 18, R = 35$ is retained and is not certified by any partition. Ordinary and optimized execution reproduce the same mathematical JSON outputs, with execution details recorded separately. These finite results and the proved sufficient criterion do not establish that every prime has a certified shell, and a shell that does not pass this strict inequality may still have a genuine target.

4 The original gate in an integral three-coordinate system

This section reconstructs the final supplied discussion of 12 September, 08:19, from its complete text. The downloaded `workbench.tex` is only a wrapper: its five mathematical input files were not recovered. The formulas below are therefore explicitly a reconstruction, not an assertion that those missing source bytes or their execution receipts have been recovered. The comparison to JT's quartic calculation is a contribution of the supplied joint investigation; the proofs here check the stated maps independently.

Fix an odd positive integer R and an integer gate g . All four marked coordinates (g, x, y, z) are retained, with g fixed by every map. On $(\mathbb{Z}/R\mathbb{Z})^3$ define

$$\begin{aligned} a_1^g(x, y, z) &= (6g + y, -6g - x - y, z - 2g + x), \\ a_2^g(x, y, z) &= (-y, x - 6g, z + 3g + y), \\ a_\infty^g(x, y, z) &= (x, y + x, z - g). \end{aligned}$$

These are the affine slices of the integral matrices in the received `es_s6_counterfactual/src/core.tex`, equation (1), whose source is the $(3, 4, \infty)$ period manuscript circulated by Levent Alpöge. Only the displayed integral representation is used; no assertion about the global diffeomorphism type of its proposed completed threefold is needed. Direct composition gives $(a_1^g)^3 = (a_2^g)^4 = a_1^g a_2^g a_\infty^g = 1$. The inverses are $(a_1^g)^2, (a_2^g)^3$ and $(x, y, z) \mapsto (x, y - x, z + g)$ respectively.

Proposition 4.1 (Integral cyclic coordinates). *For each retained g , the map*

$$t_0 = -x - y - z - 2g, \quad t_1 = -x - z + 2g, \quad t_2 = -z$$

is an affine isomorphism over $\mathbb{Z}$, hence modulo every R . Its inverse is

$$x = t_2 - t_1 + 2g, \quad y = -t_0 + t_1 - 4g, \quad z = -t_2.$$

In these coordinates the three maps become, respectively,

$$\begin{aligned} c(t) &= (t_2, t_0, t_1), \\ b_g(t) &= (t_1 - g, t_2 - g, t_0 - t_1 + t_2 + g), \\ u_g(t) &= (t_0 + t_1 - t_2 - g, t_1 + g, t_2 + g). \end{aligned}$$

Proof. The two displayed substitutions compose to the identity in all three coordinates, without a division. Substituting each a_j^g into t_0, t_1, t_2 gives the three formulas. For example $t_0(a_1^g) = -6g - y + 6g + x + y - z + 2g - x - 2g = -z = t_2$; $t_1(a_1^g) = -6g - y - z + 2g - x + 2g = t_0$ and $t_2(a_1^g) = -z + 2g - x = t_1$. This proves the first conjugacy; the second and third follow by the same explicitly given substitutions. The inverse coordinate formulas retain both g and all integer coordinates. $\square$

Proposition 4.2 (The hexagonal quotient and its exact integral inverse). *Set $X = t_2 - t_1$, $Y = t_0 - t_1$, $W = -(t_0 + t_1 + t_2)$. The image in $\mathbb{Z}^3$ is exactly*

$$\mathcal{L} = \{(X, Y, W) : W \equiv 2X - Y \pmod{3}\},$$

an index-three sublattice. Its inverse is

$$t_0 = \frac{-W - X + 2Y}{3}, \quad t_1 = \frac{-W - X - Y}{3}, \quad t_2 = \frac{-W + 2X - Y}{3}.$$

For $Q(X, Y) = X^2 - XY + Y^2$, one has $2Q(X, Y) + W^2 = 3(t_0^2 + t_1^2 + t_2^2)$. The integral operation $r(t) = (-t_1, -t_2, -t_0)$ satisfies

$$r^2 = c, \quad r^3 = -I, \quad r^6 = I, \quad r : (X, Y, W) \mapsto (X - Y, X, -W).$$

It preserves $\mathcal{L}$ and Q . It is a specified extension of the cyclic action, not claimed to be a word in the original monodromy group. The second original monodromy acts on the planar readout by $(X, Y) \mapsto (Y + 2g, -X)$ and satisfies

$$Q(Y + 2g, -X) - Q(X, Y) = 2XY + 2gX + 4gY + 4g^2.$$

Proof. Solving the three linear equations gives the displayed inverse. Its three numerators are divisible by three exactly under the one displayed congruence. The map $(X, Y, W) \mapsto W - 2X + Y \pmod{3}$ is onto, so its kernel has index three. Expanding $2(X^2 - XY + Y^2) + W^2$ after substitution cancels every cross term and leaves $3 \sum t_i^2$. Direct iteration gives the powers of r . The new readouts are $X - Y, X, -W$; their gluing residual is $-W - 2(X - Y) + X = -W - X + 2Y$, divisible by three on $\mathcal{L}$. Expanding $Q(X - Y, X)$ gives $Q(X, Y)$. Substitution into b_g gives $X' = Y + 2g, Y' = -X$; expanding the two quadratic polynomials proves the last identity, including its nonzero correction. $\square$

The readout $(t_0, t_1, t_2) \mapsto (X, Y)$ has kernel $\mathbb{Z}(1, 1, 1)$. Retaining W restores that omitted coordinate subject to the exact gluing above. Modulo three the displayed divided inverse is *not* silently used: the original integral t -coordinates remain the primary state at such levels. No unrestricted product of a plane and an integer axis replaces $\mathcal{L}$.

Proposition 4.3 (Gate order, compression and the simultaneous local return). *Put $d = \gcd(R, g)$ and $D = R/d$. For $\Delta = \langle a_1, a_2 : a_1^3 = a_2^4 = 1 \rangle$, write the above affine representation as $v \mapsto B_j v + gb_j$. Then*

$$\text{ord}(g[b_R]) = D \quad \text{in } H^1(\Delta, (\mathbb{Z}/R\mathbb{Z})_B^3).$$

Multiplication by g gives an equivariant injection from the coefficient-one system at level D to the coefficient- g system at level R :

$$(\mathbb{Z}/D\mathbb{Z})^3 \longrightarrow (\mathbb{Z}/R\mathbb{Z})^3, \quad v \longmapsto gv.$$

Its image is $(d\mathbb{Z}/R\mathbb{Z})^3$. Its inverse on the image divides representatives by d and multiplies by $(g/d)^{-1}$ modulo D . Both systems have a global fixed sheet precisely when $D = 1$. The full level- R , coefficient-zero system then has R fixed sheets, while the compressed level-one system has one.

Proof. Block multiplication gives $b_{hk} = b_h + B_h b_k$. A coboundary at the cusp has third coordinate zero because the third row of $B_\infty - I$ is zero. Third cusp evaluation is consequently a well-defined homomorphism on cohomology and sends $g[b_R]$ to $-g \pmod{R}$. If $kg[b_R] = 0$, then $R \mid kg$, equivalently $D \mid k$. Conversely $Dg = 0 \pmod{R}$ kills every cocycle coefficient, proving exact order. Since $\gcd(g/d, D) = 1$, the stated inverse proves injectivity and identifies the image. Each affine generator obeys $g(B_j v + b_j) = B_j(gv) + gb_j$, proving equivariance. For $D = 1$ every division-modulo-one means its unique zero class. A common fixed point would be fixed by the cusp; its last coordinate forces $g = 0 \pmod{R}$. When $g = 0$, the second generator forces $x = -y$ and $y = x$, so $x = y = 0$ since R is odd. Every z is then fixed by both generators. This proves the fixed-sheet counts and their existence equivalence. $\square$

In an ES leaf, $g = 4u + 1$ in channel E and $g = u + a$ in channel M. Thus the actual subgroup $\langle [g]_R \rangle$ and the subgroup generated by $g[b_R]$ are isomorphic by $kg \mapsto kg[b_R]$; minus cusp evaluation is the inverse. For $R = \prod \ell^{h_\ell}$, the CRT map is componentwise reduction of this *one* labelled state. Its inverse is $v = \sum_\ell v_\ell (R/\ell^{h_\ell})((R/\ell^{h_\ell})^{-1} \pmod{\ell^{h_\ell}} \pmod{R})$ in each coordinate. It retains the same divisor exponents, channel and gate at every factor. The order is $D = \prod_\ell \ell^{\max(h_\ell - v_\ell(g), 0)}$. Different divisors chosen separately at different primes do not define this same-leaf correspondence.

5 JT's quartic and the positive-square locus of the hexagon

Let $\eta = e^{\pi i/3}$ and $\zeta = e^{\pi i/6}$, so $\zeta^2 = \eta$. The field $\mathbb{Q}(\zeta)$ has degree four over $\mathbb{Q}$; its chosen degree-two subfield is $\mathbb{Q}(\eta)$ and the nontrivial relative automorphism sends ζ to $-\zeta$. For integers r, s ,

$$N_{\mathbb{Q}(\zeta)/\mathbb{Q}(\eta)}(r - s\zeta) = r^2 - \eta s^2, \quad N_{\mathbb{Q}(\eta)/\mathbb{Q}}(X - \eta Y) = X^2 - XY + Y^2.$$

Indeed the first product is $(r - s\zeta)(r + s\zeta)$; the second uses $\eta + \bar{\eta} = 1$ and $\eta\bar{\eta} = 1$. Consequently

$$r^4 - r^2 s^2 + s^4 = Q(r^2, s^2) = N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(r - s\zeta).$$

Multiplication by $\eta^{-1} = 1 - \eta$ sends $X - \eta Y$ to $(X - Y) - \eta X$, the exact planar hexagon map above.

Proposition 5.1 (All positive-square returns in the dihedral orbit). *Let r, s be distinct positive coprime integers. Under the group generated by $(X, Y) \mapsto (X - Y, X)$ and $(X, Y) \mapsto (Y, X)$, the points having both coordinates positive squares correspond to $(r, s), (s, r)$, and, exactly when $|r^2 - s^2| = t^2$ for a positive integer t , the two additional pairs $(\max(r, s), t), (t, \max(r, s))$. Their common quartic norm does not change. These statements retain the group element as a mark whenever the chosen path matters.*

Proof. The six rotation images of (X, Y) are

$$(X, Y), (X - Y, X), (-Y, X - Y), (-X, -Y), (Y - X, -X), (Y, Y - X).$$

The other six images are their swaps. For distinct $X, Y > 0$, the positive pairs in this list are exactly $(X, Y), (Y, X), (\max(X, Y), |X - Y|), (|X - Y|, \max(X, Y))$. Taking $X = r^2, Y = s^2$ proves the square criterion and the complete list. Any common prime divisor of t and $\max(r, s)$ would divide the other original coordinate too, contradicting $\gcd(r, s) = 1$. Rotation and swapping both preserve Q by polynomial expansion. $\square$

For $(r, s) = (3, 5)$, $25 - 9 = 16$ gives $(3, 5), (5, 3), (4, 5), (5, 4)$, all of norm $481 = 37 \cdot 13$. At $(1, 26)$ the difference is 675, not a square since $25^2 < 675 < 26^2$. The norm is $456301 = 2521 \cdot 181$, and $88^2 = 3 \cdot 2521 + 181$. Thus the exact norm bridge preserves this cofactor, which is a quadratic residue modulo 2521. A norm-changing arithmetic operation is not supplied by the hexagonal unit action. This is the precise scope of the bridge, not a conclusion that the constructions are unrelated.

6 Local maps of the completed affine covers

For this section $\mathcal{F}_D = (\mathbb{Z}/D\mathbb{Z})^3$ with the coefficient-one affine action displayed above. The base is the sphere with its three marked punctures removed, with their positively oriented monodromies a_1, a_2, a_∞ . A finite equivariant set defines its cover by taking the universal cover times the set and identifying $(\gamma z, v)$ with $(z, \gamma^{-1}v)$ under deck transformations. The marked cycles at a puncture are filled by discs with base coordinate $t = w^e$, one for each cycle of length e . This construction specifies the objects and local coordinates used in the following proof. For the cusp $a_\infty(x, y, z) = (x, y + x, z - 1)$, its k th iterate is $(x, y + kx, z - k)$; its cycle length at every point of level D is exactly D . At the finite points the lengths divide three and four by the displayed generator relations. The original level-modulus quotient alone therefore does not specify those finite local degrees.

Proposition 6.1 (Functorial modulus maps and their compactifications). *For positive odd integers $D \mid E$, coordinate reduction $\mathcal{F}_E \rightarrow \mathcal{F}_D$ is equivariant and surjective, with fibres of size $(E/D)^3$. It induces a map of the corresponding compactified covers. At a marked source branch cycle of length e_E mapping to a target branch cycle of length e_D , one has $e_D \mid e_E$, and the local degree is e_E/e_D . At the cusp this is E/D .*

For coprime positive odd integers D, E , coordinate CRT gives an equivariant bijection

$$\mathcal{F}_{DE} \xrightarrow{\sim} \mathcal{F}_D \times \mathcal{F}_E,$$

with coordinate inverse $a + D((b - a)D^{-1} \bmod E) \bmod DE$. The corresponding unramified cover is the fibre product over the punctured base. Its compactification is the normalization of the fibre product of the compactified covers over $\mathbb{P}^1$. The cocycles and local systems have the induced reduction and CRT maps. Every product and every map retains the same base and candidate label.

Proof. All three affine formulas have integral coefficients, hence commute with coordinate reduction. Each residue modulo D has exactly E/D lifts modulo E , proving the fibre count in three coordinates. Equivariance gives the map of covers away from the punctures.

At a fixed puncture, let e_E be the source cycle length and e_D its image cycle length. The e_E -th power of target monodromy fixes the image sheet, so $e_D \mid e_E$. Choose compatible sheet origins and local coordinates with base parameter

$$t = w_E^{e_E} = w_D^{e_D}.$$

The prescribed punctured-cover map has the local form

$$w_D = \zeta w_E^{e_E/e_D}, \quad \zeta^{e_D} = 1,$$

where the root of unity retains the target branch mark. The positive integer exponent proves holomorphic extension, unique by agreement on the punctured disc. At infinity $e_E = E$ and $e_D = D$ by the cusp-cycle theorem. At the finite branch points their values are the actual cycle lengths dividing three and four; they need not equal the moduli.

The displayed CRT formula is inverse to both coordinate reductions. It commutes with all three monodromies, so it identifies the unramified level- DE cover with the fibre product of the two unramified covers. To complete this identification, consider source and target branch cycles of lengths e_D, e_E . Their local fibre product has equation

$$x^{e_D} = y^{e_E}.$$

Put $d = \gcd(e_D, e_E)$, $r = e_D/d$, $s = e_E/d$. Its d branches are $x^r = \xi y^s$ for $\xi^d = 1$. On a branch, choose $\eta^r = \xi$. The normalization has the parametrization

$$x = \eta w^s, \quad y = w^r, \quad t = w^{\text{lcm}(e_D, e_E)}.$$

The two maps to the base agree because $\eta^{e_D} = \xi^d = 1$. Since $\gcd(r, s) = 1$, Bezout's identity gives a birational inverse in the field of meromorphic germs, and the parameter disc is smooth. The product monodromy has precisely d cycles of length $\text{lcm}(e_D, e_E)$: a simultaneous iterate returns exactly when divisible by both lengths, and the number of orbits is $e_D e_E / \text{lcm}(e_D, e_E) = d$. Thus the normalized local branches are exactly the completed product cover. Their unique extensions glue to the compactified level- DE cover. This argument is componentwise for disconnected covers and keeps every candidate label and marked branch.

For example, reduction from level 15 to level 3 sends the a_1 cycle

$$(0, 0, 0) \mapsto (6, 9, 13) \mapsto (0, 9, 2) \mapsto (0, 0, 0)$$

to the cycle

$$(0, 0, 0) \mapsto (0, 0, 1) \mapsto (0, 0, 2) \mapsto (0, 0, 0).$$

The local degree here is $3/3 = 1$, whereas at the cusp it is $15/3 = 5$. For the coprime levels three and five the cusp fibre product $x^3 = y^5$ is singular; its normalization is $x = w^5, y = w^3, t = w^{15}$. This records why normalization is part of the compactified product statement. $\square$

7 The marked cubic-surface contraction and the toric hexagon

This section reconstructs the explicit geometric comparison in the supplied discussion, lines 2017–2055. All point labels and directions below are part of the objects. The split degree-six model and its six boundary curves occur in [Mark Blunk, *Del Pezzo surfaces of degree 6 over an arbitrary field*, §2, Proposition 2.1, printed p. 3](#); the Picard maps, signs and torus coordinates needed for the present comparison are proved here.

7.1 The marked surfaces, contraction and complete inverse fibres

Work over $\mathbb{C}$. Let $p_1, \dots, p_6$ be six distinct marked points of $\mathbb{P}^2$, no three collinear and not all six on a conic. Choose coordinates with

$$p_1 = [1 : 0 : 0], \quad p_2 = [0 : 1 : 0], \quad p_3 = [0 : 0 : 1].$$

This is a nonempty explicitly realized domain: one choice is

$$p_4 = [1 : 1 : 1], \quad p_5 = [1 : 2 : 3], \quad p_6 = [1 : 3 : 2]. \quad (\text{PH1})$$

For this choice the determinants of the twenty point triples, in lexicographic index order, are

$$1, 1, 3, 2, -1, -2, -3, 1, -1, -5, 1, 1, 1, -2, -1, 1, 1, 2, 1, -3,$$

with the points as rows in increasing label order; in particular they are all nonzero. A conic through the first three points has the form $Ax_0x_1 + Bx_0x_2 + Cx_1x_2 = 0$. Its values at the last three have coefficient matrix

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 6 \\ 3 & 2 & 6 \end{pmatrix}, \quad \det = 7,$$

so no conic passes through all six.

Put

$$Y = \text{Bl}_{p_1, p_2, p_3} \mathbb{P}^2, \quad S = \text{Bl}_{p_1, \dots, p_6} \mathbb{P}^2.$$

Let $q_a \in Y$ be the unchanged point above p_a , for $a = 4, 5, 6$. All three lie in the complement of the coordinate triangle and its exceptional divisors. The morphism

$$f : S = \text{Bl}_{q_4, q_5, q_6} Y \longrightarrow Y \quad (\text{PH2})$$

contracts exactly the three exceptional curves E_4, E_5, E_6 . Denote the line pullback and exceptional classes on S by $H, E_1, \dots, E_6$, and those on Y by h, e_1, e_2, e_3 .

Here are the fibres and their inverse data explicitly. Near a centre q_a take regular local coordinates (ξ, η) vanishing there. The blowup is

$$\{(\xi, \eta, [s : t]) : \xi t = \eta s\}.$$

Its two charts are (ξ, t) with $\eta = \xi t$, and (s, η) with $\xi = s\eta$. Over the centre the fibre is the whole $\mathbb{P}^1$ of directions $[s : t]$; elsewhere the unique direction is $[\xi : \eta]$. Thus f is an isomorphism

away from the three specified fibres, and retaining the centres and these exceptional directions gives the full blowup return. No inverse at a contracted point discards its direction.

The Picard groups and intersection forms are

$$\mathrm{Pic}(S) = \mathbb{Z}H \oplus \bigoplus_{i=1}^6 \mathbb{Z}E_i, \quad H^2 = 1, \quad HE_i = 0, \quad E_iE_j = -\delta_{ij},$$

and the analogous basis (h, e_1, e_2, e_3) on Y . To see the additional summand at a blowup, push a divisor to the smooth base and subtract its total transform; the difference is supported on the exceptional curve. Its coefficient is unique since that curve has self-intersection -1 , as follows from the transition of its normal coordinate between the two charts above. Applying this argument successively gives the displayed bases. The maps are exactly

$$f^*h = H, \quad f^*e_i = E_i \quad (1 \leq i \leq 3), \quad f_*H = h, \quad f_*E_i = e_i \quad (i \leq 3), \quad f_*E_a = 0 \quad (a \geq 4). \quad (\text{PH3})$$

Hence $f_*f^* = 1$, the kernel of f_* is $\mathbb{Z}E_4 \oplus \mathbb{Z}E_5 \oplus \mathbb{Z}E_6$, and it is orthogonal to $f^*\mathrm{Pic}(Y)$. For $D = dH - \sum m_i E_i$, $D' = d'H - \sum m'_i E_i$, one has

$$f^*f_*D = D + \sum_{a=4}^6 m_a E_a, \quad (f_*D)(f_*D') = DD' + \sum_{a=4}^6 m_a m'_a. \quad (\text{PH4})$$

These identities prove both the precise lost sublattice and the correction to intersection numbers. The canonical classes obey

$$-K_S = 3H - \sum_{i=1}^6 E_i = f^*(-K_Y) - E_4 - E_5 - E_6, \quad -K_Y = 3h - e_1 - e_2 - e_3, \quad K_S^2 = 3, \quad K_Y^2 = 6. \quad (\text{PH5})$$

The canonical-class formula follows in a blowup chart from $d\xi \wedge d\eta = \xi d\xi \wedge dt$ and then from the canonical class $-3h$ of the plane.

For completeness, these marked blowups give the stated cubic model. The cubic forms through the six points have dimension four: their six evaluation conditions are independent, since a conic through any five points, multiplied by a line avoiding the sixth, separates that sixth evaluation. Such a conic exists, is unique by Bezout, and is irreducible because a pair of lines contains at most four of the points. The cubics have no further base point: for a point outside the six, the pairs whose joining line contains it form a matching of at most three edges. There are fifteen perfect matchings of six labels, and each forbidden edge belongs to three; hence at least six perfect matchings avoid all forbidden edges. Such a matching supplies a product of three joining lines nonzero there. At each marked point, matchings with two different partners give independent first-order directions. Thus the induced anticanonical system is base-point free on S . It is ample as well. For a nonexceptional irreducible curve of degree d and multiplicities m_i , distinct from the six conics above, Bezout with each such conic gives $2d \geq \sum_{j \neq i} m_j$. Summing gives $\sum m_i \leq 12d/5 < 3d$, so $(-K_S)C > 0$. Each of the six conics and every E_i has anticanonical degree one. Together with $K_S^2 = 3$, the Nakai–Moishezon criterion proves ampleness. The resulting morphism to $\mathbb{P}^3$ is finite, and its nondegenerate surface image has degree at least two. The equality $3 = \deg(\text{morphism}) \deg(\text{image})$ makes it birational onto a cubic surface. A general hyperplane pullback is a smooth curve of genus one by Bertini and adjunction. Its plane cubic image has arithmetic genus one, so it is smooth; hence the cubic surface has no singular curve. A hypersurface with no codimension-one singularity is normal. The finite birational morphism to this normal surface is an isomorphism. In particular the anticanonical model is a smooth cubic surface.

7.2 All twenty-seven line classes and their contracted images

The complete list on S is

$$E_i \ (1 \leq i \leq 6), \quad L_{ij} = H - E_i - E_j \ (i < j), \quad Q_i = 2H - \sum_{j \neq i} E_j \ (1 \leq i \leq 6). \quad (\text{PH6})$$

The last two classes are the strict transforms of the actual joining lines and the unique conics through the indicated five points. Their curves are smooth rational, have square -1 , and anticanonical degree one, so their images in the cubic model are lines. Conversely a line is a smooth rational curve with square -1 by adjunction. Unless exceptional, it has degree $d \geq 1$ and nonnegative multiplicities satisfying

$$\sum m_i = 3d - 1, \quad \sum m_i^2 = d^2 + 1.$$

Cauchy gives $(3d - 1)^2 \leq 6(d^2 + 1)$, hence $d \leq 2$. At $d = 1$ exactly two multiplicities are one; at $d = 2$ exactly five are one. This proves (PH6) is exhaustive without omitting a line class.

Let $J = \{1, 2, 3\}$ and $K = \{4, 5, 6\}$. The entire contraction correspondence is the following table. A zero divisor class in its second row records a contracted curve, not an inverse point of a divisor map.

curve on S	image class on Y	image curve square
$E_i \ (i \in J)$	e_i	-1
$E_a \ (a \in K)$	$0 \ (\text{point } q_a)$	not a curve
$L_{ij} \ (i, j \in J)$	$h - e_i - e_j$	-1
$L_{ia} \ (i \in J, a \in K)$	$h - e_i$	0
$L_{ab} \ (a, b \in K)$	h	1
$Q_i \ (i \in J)$	$2h - \sum_{j \in J \setminus \{i\}} e_j$	2
$Q_a \ (a \in K)$	$2h - e_1 - e_2 - e_3$	1

(PH7)

Each noncontracted entry is the actual line or conic with its remaining marked incidence conditions; coincident divisor classes do not identify distinct image curves. The total-transform inverse is

$$f^* f_* L_{ij} = L_{ij} + \sum_{a \in K \cap \{i, j\}} E_a, \quad f^* f_* Q_i = Q_i + \sum_{a \in K \setminus \{i\}} E_a, \quad (\text{PH8})$$

because the original plane curves pass simply through exactly those centres. Equations (PH3)–(PH4) prove the table and every square in it.

The six curves

$$E_1, L_{12}, E_2, L_{23}, E_3, L_{13} \quad (\text{PH9})$$

avoid the three contraction centres and return isomorphically to their six namesakes on Y . Each has square -1 . The intersection rules give $E_i L_{jk} = 1$ exactly when $i \in \{j, k\}$; different E_i are disjoint, and the three displayed L_{ij} are pairwise disjoint. Thus (PH9) is exactly the cycle with neighbouring intersection one and every other intersection zero. Its sum is $-K_Y$.

7.3 A six-cycle realized by Cremona and a marked configuration map

Put $\alpha = H - E_1 - E_2 - E_3$, so $\alpha^2 = -2$ and $K_S \alpha = 0$. Define the reflection and permutation

$$C(D) = D + (D\alpha)\alpha, \quad \sigma = (1 \ 3 \ 2), \quad \sigma(H) = H, \quad \sigma(E_i) = E_{\sigma(i)},$$

where σ fixes the labels in K . Their composite is $T = \sigma C$. The two operations commute, $C^2 = 1$, and $\sigma^3 = 1$. Expansion using $\alpha^2 = -2$ proves $C(D)C(D') = DD'$, and both operations

fix K_S . Consequently T is an integral intersection isometry, fixes the canonical class, and $T^3 = C$, $T^6 = 1$. Its exact basis action is

$$\begin{aligned} T(H) &= 2H - E_1 - E_2 - E_3, \\ T(E_1) &= L_{12}, \quad T(E_2) = L_{23}, \quad T(E_3) = L_{13}, \quad T(E_a) = E_a \quad (a \in K). \end{aligned} \quad (\text{PH10})$$

In particular it sends the six entries of (PH9) successively to the next, including $L_{13} \mapsto E_1$, and its order is exactly six. Its full permutation of all twenty-seven classes is

$$\begin{aligned} &(E_1 \ L_{12} \ E_2 \ L_{23} \ E_3 \ L_{13}), \quad (E_4)(E_5)(E_6), \\ &(L_{14} \ L_{34} \ L_{24}) \ (L_{15} \ L_{35} \ L_{25}) \ (L_{16} \ L_{36} \ L_{26}), \quad (Q_1 \ Q_3 \ Q_2), \\ &(L_{45} \ Q_6) \ (L_{46} \ Q_5) \ (L_{56} \ Q_4). \end{aligned} \quad (\text{PH11})$$

To verify completeness directly, a mixed class L_{ia} has $L_{ia}\alpha = 0$, so maps to $L_{\sigma(i),a}$; a class L_{ab} in K has product one with α , so maps to Q_c for $K \setminus \{a, b\} = \{c\}$; the latter has product -1 and maps back. A Q_i with $i \in J$ has product zero and maps to $Q_{\sigma(i)}$. Together with (PH10), these calculations prove every entry of (PH11); intersection preservation holds for every pair, not only for neighbouring sides of the hexagon.

This is realized geometrically on the degree-six surface. The quadratic plane Cremona map and the coordinate permutation are

$$\kappa[x_0 : x_1 : x_2] = [x_1x_2 : x_0x_2 : x_0x_1], \quad \rho[x_0 : x_1 : x_2] = [x_1 : x_2 : x_0].$$

Their composite is

$$F = \rho\kappa : [x_0 : x_1 : x_2] \dashrightarrow [x_0x_2 : x_0x_1 : x_1x_2]. \quad (\text{PH12})$$

The blowup Y is also the graph closure

$$\{([x], [y]) \in \mathbb{P}^2 \times \mathbb{P}^2 : x_0y_0 = x_1y_1 = x_2y_2\}.$$

Away from the three coordinate points the second coordinate is the displayed Cremona value; at each point the two blowup charts give its exceptional direction. Thus switching the two projective factors extends κ to an automorphism of Y , and simultaneous permutation of the coordinates extends ρ . Their action on curve classes is the *pushforward* $T = \sigma C$; the pullback is $T^{-1} = C\sigma^{-1}$. For instance a general line has Cremona transform a conic through the three centres, and E_i maps to the opposite joining line, proving (PH10) directly.

On a six-point blowup the exact geometric statement retains the changed centres. Set $q'_a = F(q_a)$ and $S' = \text{Bl}_{q'_4, q'_5, q'_6} Y$. Then F lifts to an isomorphism $\tilde{F} : S \rightarrow S'$ making $f'\tilde{F} = Ff$, with the exceptional label a sent to the exceptional label a at q'_a . Its inverse is the lift of F^{-1} . On exceptional directions its map is the projectivized invertible derivative $\mathbb{P}(dF_{q_a})$, obtained by substituting the two local blowup charts. For (PH1), the new plane centres are exactly

$$p'_4 = [1 : 1 : 1], \quad p'_5 = [3 : 2 : 6], \quad p'_6 = [2 : 3 : 6].$$

Thus (PH10)–(PH11) are the complete marked pushforward between these configurations; they do not assert an automorphism of the fixed generic six-point configuration. The contraction maps commute with T on their Picard groups because the kernel spanned by E_4, E_5, E_6 is fixed.

7.4 Torus coordinates, the exact fan, and the cusp-side convention

Use the following specified coordinates on the dense torus of Y :

$$u = x_2/x_0, \quad v = x_2/x_1, \quad [x_0 : x_1 : x_2] = [v : u : uv]. \quad (\text{PH13})$$

Substitution in (PH12) gives the genuine torus automorphism

$$F(u, v) = (u/v, u), \quad F^{-1}(u, v) = (v, v/u), \quad F^3(u, v) = (u^{-1}, v^{-1}), \quad F^6(u, v) = (u, v). \quad (\text{PH14})$$

For example $F^2(u, v) = (v^{-1}, u/v)$, proving the third-power formula by one more substitution. At a retained centre (u, v) the exceptional direction map in (PH2) is completely explicit:

$$dF_{(u,v)} = \begin{pmatrix} 1/v & -u/v^2 \\ 1 & 0 \end{pmatrix}, \quad \det dF = u/v^2, \quad [P : Q] \mapsto [vP - uQ : v^2P]. \quad (\text{PH14a})$$

At the image centre $(U, V) = (u/v, u)$ its inverse sends $[P' : Q']$ to $[U^2Q' : UQ' - VP']$. Substitution gives $[u^2P : u^2Q] = [P : Q]$, proving the inverse on the entire exceptional $\mathbb{P}^1$, including vertical and infinite slope directions. The valuation vectors $(\text{ord}_B u, \text{ord}_B v)$ of the boundary curves in the retained order are precisely

B	E_1	L_{12}	E_2	L_{23}	E_3	L_{13}
$\text{ord}_B u$	1	1	0	-1	-1	0
$\text{ord}_B v$	0	1	1	0	-1	-1

(PH15)

For a joining line these are its zero and pole orders in (PH13). At E_1 , both x_1 and x_2 vanish to first order while x_0 is a unit, giving $(1, 0)$; the other two exceptional rows follow in the same way from the displayed coordinate ratios. Thus the six rays, in the exact curve order, are

$$n_1 = (1, 0), \quad n_2 = (1, 1), \quad n_3 = (0, 1), \quad n_4 = (-1, 0), \quad n_5 = (-1, -1), \quad n_6 = (0, -1).$$

Adjacent pairs have determinant one. The cones between them are the six smooth toric charts of the three-point blowup: the original plane fan's three cones are each subdivided by the sum of their two generating rays, which is the blowup operation in the local charts of (PH2). On this cocharacter lattice (PH14) acts by

$$R_6 = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_6 n_i = n_{i+1}, \quad R_6^3 = -I,$$

with subscripts modulo six. Its action on characters is the transpose. The equality $Q(X - Y, X) = Q(X, Y)$ for $Q = X^2 - XY + Y^2$ proves that these are the same six norm-one vectors as in the integral hexagonal quotient. Because R_6 carries every cone to the next cone, its monomial map and inverse carry the dual-cone monomial coordinate rings into the corresponding chart rings. This proves extension across every boundary chart directly, in agreement with the graph model above.

The source cusp marks its rays, in §9.3 of the [circulated period manuscript](#), by

$$(1, 0), (1, -1), (0, -1), (-1, 0), (-1, 1), (0, 1).$$

The integral matrix $B = \text{diag}(1, -1)$ sends each n_i to the corresponding source ray, preserves opposite pairs, and is its own inverse. It induces the exact torus coordinate change $(u, v) \mapsto (u, v^{-1})$. In that convention the conjugate rotation is $BR_6B = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$ and the conjugate torus map is $(u, v) \mapsto (uv, u^{-1})$.

This identifies the marked smooth toric normalizations and their six curves. The source further supplies a normalization map $\nu : dP_6 \rightarrow W$ identifying each side with its opposite. With that source map retained, the actual comparison is the composite $S \xrightarrow{f} Y \xrightarrow{B} dP_6 \xrightarrow{\nu} W$. The first map has exactly the three $\mathbb{P}^1$ fibres proved above. The last has one preimage off the boundary, two branches at a general double-curve point, and three at each triple point, as its local equations $xy = 0$ and $xyz = 0$ show. The source marks the two triple-point preimages by the alternating vertex sets $\{p_{12}, p_{34}, p_{56}\}$ and $\{p_{23}, p_{45}, p_{61}\}$. All side-identification maps remain attached to ν ; the fan map alone does not reconstruct their gluing parameters or identify the singular quotient with its smooth normalization. No global six-sphere claim is used in the Picard, torus or contraction proofs.

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